

RESEARCH ARTICLE SUMMARY

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Morphoelectric and transcriptomic divergence of the layer 1 interneuron repertoire in human versus mouse neocortex

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INTRODUCTION: The neocortex contributes to higher-order cognitive processes in part by generating predictions about the external world and comparing them with sensory information. This process is mediated by the complex architecture and circuitry of the neocortex, which in most cortical areas consists of six layers. The outermost layer, layer 1 (L1), is thought to be where internally generated predictions are compared with sensory signals via a confluence of long-range feedback axons and

local-neuron dendrites. Local inhibitory interneurons complete the L1 architecture and play a strong modulating role in these feedback connections.

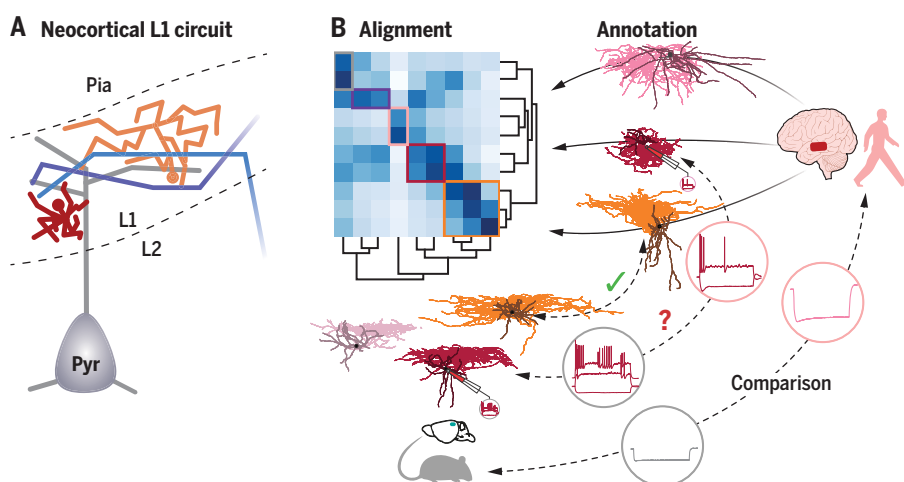
RATIONALE: Studies in rodents have shown that different types of L1 interneurons perform different functions. Distinct cell types have also been noted in human L1, but their function and gene expression have not been fully characterized. Single-cell genomics is revolutioniz-

ing the study of cellular diversity in the brain through the construction of transcriptomic cell-type taxonomies. As described in companion papers, these methods can capture the exceptional diversity of the brain and allow comparative analysis across species, providing a framework to bound the problem of cellular diversity. However, this framework remains disconnected from cellular function without simultaneous measurement of physiology, morphology, and gene expression (known as “Patch-seq”), annotating these cell types with their functional properties. We applied Patch-seq with brain tissue from human neurosurgical patients to reveal cell-type diversity in human neocortical L1 and to make principled cross-species comparisons.

RESULTS: We leveraged single-cell transcriptomics to define neuronal cell types in human and mouse L1 and to quantitatively identify homologous subclasses across species. Although established marker genes for L1 cell types were not well conserved from mouse to human, whole-transcriptome analysis enabled cross-species alignment. We observed differences in the abundance of L1 cell types, including cell types found in human but not mouse L1, and stronger differences in gene expression between human types than between mouse types.

Analysis of Patch-seq data showed that L1 cell types in both human and mouse were differentiated by morphological and electrophysiological features related to basic neuronal function. However, the features that distinguish cell types were different in mouse versus human L1. We highlighted and characterized two cell types in human L1 with distinctive phenotypes unmatched by their putative mouse homologues: the “rosehip” cell type with compact axonal structure and large rosehip-like pre-synaptic boutons and a larger PAX6-expressing type that fired high-frequency action potential bursts at the onset of stimulation.

CONCLUSION: Our analysis provides tools for understanding human L1 diversity and suggests hypotheses and paths for further investigation. The observed cross-species differences suggest differential regulation of higher-order input to the neocortical circuit in human and mouse. We propose that the dramatic expansion in primates of deeper layers of neocortex and the increased cellular diversity that accompanied it necessitated new types of inhibition in L1. ■



Innovations in human layer 1 interneurons. (A) L1 consists of long-range axons (blue), pyramidal neuron dendrites, and diverse GABAergic interneurons. (B) Aligning transcriptomic cell-type taxonomies between human and mouse defined putative homologous L1 interneuron subclasses. Patch-seq experiments in brain slices from neurosurgical patients annotated transcriptomic cell types with morphoelectric properties. Cross-species comparison revealed clear matches, divergence of specific subclasses, and overall shifts in properties.

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RESEARCH ARTICLE

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Morphoelectric and transcriptomic divergence of the layer 1 interneuron repertoire in human versus mouse neocortex

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Neocortical layer 1 (L1) is a site of convergence between pyramidal-neuron dendrites and feedback axons where local inhibitory signaling can profoundly shape cortical processing. Evolutionary expansion of human neocortex is marked by distinctive pyramidal neurons with extensive L1 branching, but whether L1 interneurons are similarly diverse is underexplored. Using Patch-seq recordings from human neurosurgical tissue, we identified four transcriptomic subclasses with mouse L1 homologs, along with distinct subtypes and types unmatched in mouse L1. Subclass and subtype comparisons showed stronger transcriptomic differences in human L1 and were correlated with strong morphoelectric variability along dimensions distinct from mouse L1 variability. Accompanied by greater layer thickness and other cytoarchitecture changes, these findings suggest that L1 has diverged in evolution, reflecting the demands of regulating the expanded human neocortical circuit.

Neocortical layer 1 (L1) is implicated in several higher-order brain functions, including state modulation (1), learning (2–5), sensory perception (6), and consciousness (7). The neural circuitry that mediates these functions consists of converging pyramidal-cell dendrites; long-range axons originating from thalamic, cortical, and neuro-

modulatory regions; and axons from local γ -aminobutyric acid-producing (GABAergic) interneurons (8). Although some of this inhibitory input arises from other layers (Martinotti cells, for example), much of it arises from neurons with cell bodies in L1, an entirely GABAergic cell population with distinct developmental origins (9, 10). Emerging evidence suggests that these L1 interneurons profoundly shape cortical processing and that diversity within this population is linked to diversity of function (11, 12). As such, the L1 interneuron repertoire is a potential site of evolutionary divergence that could contribute to specialized cortical function in humans and other primates. In rodents, a progression of classification schemes for L1 neurons (13–18) has evolved toward a view of four canonical types based on molecular markers (19), but the robustness of this scheme, both across modalities and across species, remains unclear (particularly in human and nonhuman primates). Indeed, the observation of a “rosehip” cell type found in human and not mouse neocortex (20) highlights the

importance of studying human L1 to identify potential species specializations and to relate mouse literature to human L1 cell types and function.

Traditionally, L1 cell types have been defined by their morphology, sublamina location, intrinsic membrane properties, and a handful of marker genes. However, applying distinctive features from rodent to define and study human cell types can be tenuous. Instead, single-cell whole-transcriptome data can be leveraged to define cross-species cell-type homologies (21, 22) and reveal genetic and phenotypic diversity obscured by the marker gene approach (23, 24), as observed in vivo in mouse L1 (11). The Patch-seq technique (25, 26), which combines patch-clamp electrophysiology, RNA sequencing, and morphological reconstruction from the same neuron, gives us unprecedented ability to reveal cell-type diversity in human L1. We leveraged this multimodal data to provide a comprehensive view of cell-type distinctions previously proposed from a subset of modalities, make principled cross-species comparisons, and robustly identify distinct phenotypes found in human L1 across modalities.

Results

L1 Patch-seq pipeline and transcriptomic references

To guide our analysis of L1 cell types, we used transcriptomic types (t-types) previously defined in reference datasets from human middle temporal gyrus (MTG) and mouse primary visual cortex (V1Sp), single-nucleus RNA sequencing (snRNA-seq) in human, and single-cell RNA-seq (scRNA-seq) in mouse (21, 27). With annotations from layer dissections as a guide, we identified 10 L1 t-types in human and 8 in mouse (Materials and methods and fig. S1A). In uniform manifold approximation and projection (UMAP) (28) views of transcriptomic data (Fig. 1A), many human t-types formed separated clusters, with others clustered in groups of related t-types, whereas mouse L1 t-types showed more continuous variability. This contrast suggests stronger transcriptomic specialization in human L1, similar to that found in supragranular excitatory neurons (29), and indicates that more robust groupings of L1 types into highly distinct transcriptomic subclasses can be delineated in human. We grouped related human t-types into L1-focused transcriptomic subclasses by quantifying the pairwise distinctness of t-types in terms of a d' separation of likelihoods (23, 24). This grouping formed three subclasses, with three t-types remaining ungrouped (Fig. 1B). Expression of the inhibitory subclass markers *PAX6* and *LAMP5* (27, 30) and the t-type marker *MC4R* also closely matched these subclass boundaries (Fig. 1C).

Human subclass marker genes did not clearly identify subclasses in mouse, posing a challenge

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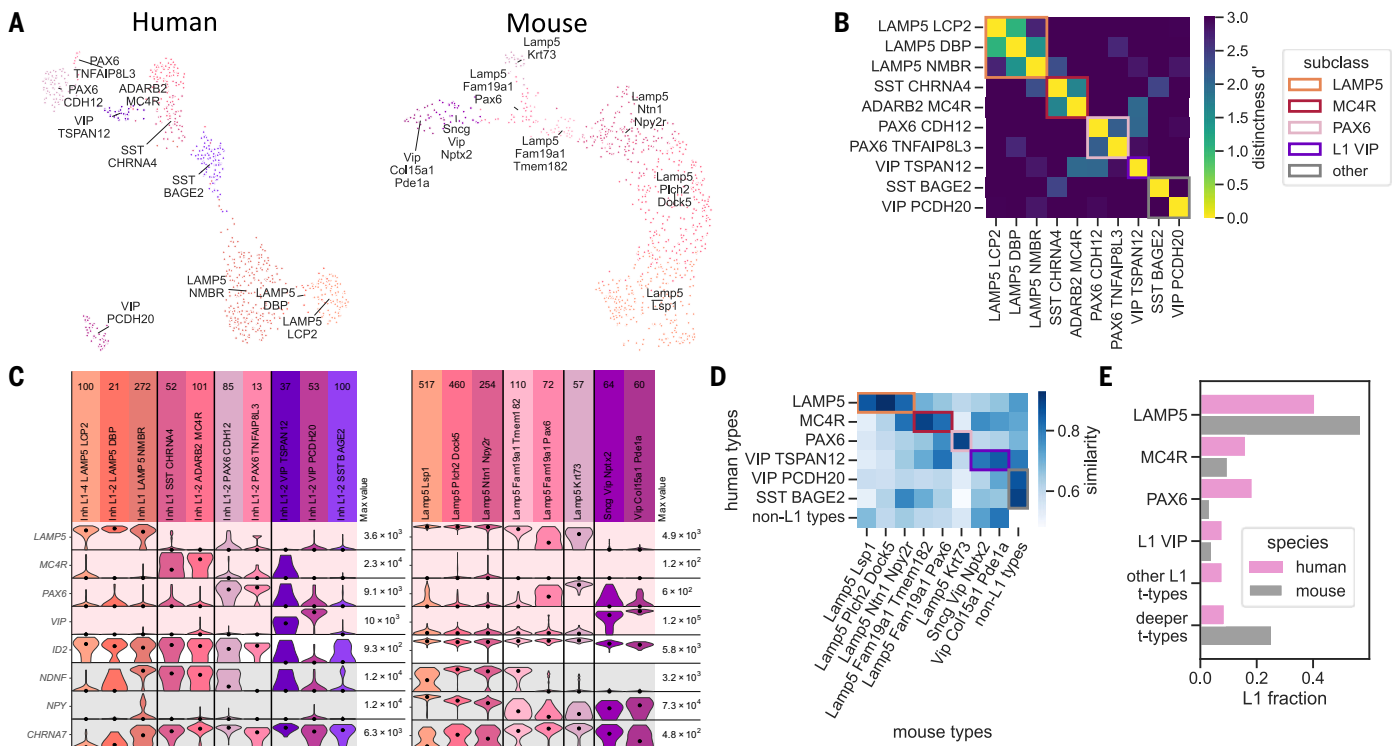


Fig. 1. snRNA-seq demonstrates L1 diversity and provides a reference for Patch-seq transcriptomic mapping. (A) UMAP projections of (left) human and (right) mouse gene expression for L1 t-types (single-neuron or -nucleus RNA-seq).

(B) Human t-types grouped by thresholding transcriptomic distinctness d' , defining subclasses. Remaining ungrouped t-types are marked as “L1 VIP” or “other” according to cross-species homology results. (C) Expression of canonical and t-type-specific marker genes across L1 t-types in (left) human and (right) mouse. Pink background indicates human subclass markers; gray background indicates classical mouse markers. Vertical lines group t-types by subclass. Violin plots show normalized probability density of gene expression (shape width) and median expression (dots), with expression in $\log(\text{CPM}+1)$ normalized by gene

for each species and maximal expression in counts per million (CPM) noted at right. (D) Mouse t-types grouped with human t-types into homologous subclasses (outlined) by thresholding similarity scores (heatmap intensity, from cluster overlap in integrated transcriptomic space). Non-L1 t-types are excluded, with maximal similarity over all non-L1 types shown for reference. (E) Proportions of subclasses and unclassified t-types in L1 Patch-seq data, by species. “Other L1 t-types” refers to t-types in human L1 with no mouse homolog in L1. “Deeper t-types” refers to types found in L1 in lower proportions, not meeting the criteria for core L1 t-types. All cross-species proportion differences (except L1 VIP) significant at FDR-corrected $P < 0.001$, one-versus-rest Fisher’s exact tests.

for cross-species comparison. Marker genes were either not expressed in any mouse L1 type (MC4R) or were expressed broadly and overlapped with other markers (LAMP5) (Fig. 1C, top). Similarly, markers previously suggested for L1 subclasses in mouse (19) showed graded or complete lack of expression in human L1 (Fig. 1C, bottom). Notably, the marker *Id2*, suggested for distinguishing a class of interneurons including all L1 types (*Pvalb*[−]/*Sst*[−]/*Vip*[−]) (31), was consistently expressed across most L1 types in both species (with the exception of marginal expression in ungrouped human t-types).

Given the lack of conserved subclass markers across species, we instead grouped mouse t-types for cross-species analysis by using cluster distances in an integrated transcriptomic space (21) (fig. S1C), identifying each mouse t-type with the most similar human subclass or ungrouped t-type (Fig. 1D and fig. S1B). These matches resulted in four homology-driven subclasses (called subclasses hereafter) with proportions largely comparable across species (Fig. 1E; PAX6 is the notable exception),

named by the subclass marker genes in human. Two additional human L1 t-types (SST BAGE2 and VIP PCDH20) were excluded from cross-species L1 subclasses because of their homology to t-types found in deeper layers but not in L1 of mouse neocortex. This observation suggests that there is a shift in some of the interneuron diversity across laminar boundaries between mouse and human (32). Reinforcing the validity of these subclass divisions in mouse, we noted likely matches to previous mouse L1 subclasses (19) based on marker-gene expression (Fig. 1C and table S1): neurogliaform (NGFC) cells (*Npy*[−]/*Ndnf*[−]) to LAMP5, canopy cells (*Npy*[−]/*Ndnf*[−]) to MC4R, and $\alpha 7$ cells (*Ndnf*[−]/*Vip*[−]/*Chrnat*[−]) to PAX6 (11). However, uncertainty in these matches highlights the need for further confirmation based on morpho-electric properties.

To characterize morphoelectric and transcriptomic diversity across human L1 cell types, we used a previously established pipeline for high-throughput data acquisition and analysis (26, 29) to generate and release a comprehensive L1

Patch-seq dataset. Human tissue was obtained from surgical samples and processed with standardized protocols; most samples originated from the MTG, along with smaller fractions from other temporal and frontal areas (data S1). All cells were filtered for transcriptomic ($n = 250$) and electrophysiological quality ($n = 194$), and a subset of neurons ($n = 71$) with sufficient cell labeling was imaged at high resolution and their dendritic and axonal morphologies reconstructed.

We assigned transcriptomic cell types and subclasses to Patch-seq samples by using a “tree mapping” classifier, a decision tree based on the transcriptomic taxonomy structure (Materials and methods) (24). Validating these assignments, we visualized t-type labels from Patch-seq and reference datasets in a joint UMAP projection by using alignment methods from the Seurat package (33) and found strong correspondence (fig. S1E). Additionally, marker genes used by the classifier showed strong correlation by t-type between Patch-seq data and the snRNA-seq reference (Fig. S1D).

Because Patch-seq sampling was not uniform across cortical layers, we also measured the laminar distribution of L1 t-types with a spatially resolved, robust, and reliable single-cell profiling technique [multiplexed error-robust fluorescence in situ hybridization (MERFISH) (32)] (Materials and methods). MERFISH laminar distributions were compatible with those from layer dissections of snRNA-seq, confirming that human L1 t-types are predominantly found in L1 or on the L1/L2 border and demonstrating t-type-specific distributions across deeper layers and within L1 for certain types (fig. S2, B to D). Proportions of t-types within L1 were also generally matched between Patch-seq, snRNA-seq, and MERFISH (fig. S2A), with one exception: Patch-seq had fewer SST BAGE2 cells and more PAX6 CDH12 and VIP TSPAN12 cells compared with snRNA-seq [$P < 0.05$, false discovery rate (FDR)-corrected Fisher's exact test]. MERFISH had intermediate proportions of SST BAGE2, with no significant differences compared with Patch-seq ($P > 0.18$), perhaps indicating differences caused by technical factors in snRNA-seq only, such as imprecise layer dissections.

Morphoelectric diversity in human L1

Organizing the Patch-seq dataset by transcriptional subclass revealed the exceptionally diverse morphology and physiology of human L1 interneurons. Morphologically, subclasses were distinguished by vertical orientation of axons and dendrites, axon extent and shape, and dendrite branching (Fig. 2, A and D; fig. S4; and data S2). Electrophysiologically, subclasses were distinguished by subthreshold properties such as the ratio of the steady-state to the transient peak of hyperpolarization (sag), as well as several suprathreshold properties, including firing rate, single-action-potential kinetics, and adaptation of spike kinetics during trains of action potentials (Fig. 2, B and C, and data S2). Spike adaptation properties showed a strong inverse relationship with sag across the dataset (fig. S3A). Sag is often mediated by hyperpolarization-activated, cation-nonselective (HCN) channels (34) and spike broadening by specific K^+ channels (35, 36), so this finding may indicate a functional relationship between these channels in all subclasses of human L1 neurons.

LAMP5 cells, the largest subclass, corresponded to the classical NGFC cell type (37), with highly branched, descending dendrites and horizontally elongated axons, with either a rectangular or triangular shape (Fig. 2A). Their electrophysiological phenotype was relatively undistinguished, with firing-rate adaptation and sag present but small (Fig. 2B). PAX6 cells had axons similar to those of LAMP5 cells, occasionally with descending branches, and sparse downward dendrites, along with minimal sag and high initial firing rate at the onset of response to depolarizing current injection

(Fig. 2, A and B). MC4R cells had extremely compact, ball-like axonal arbors, along with strong sag; on this basis, they were tentatively identified as a match to the recently discovered rosehip type (further characterized in the discussion of distinctive neuronal phenotypes below) (20). L1 VIP (TSPAN12 t-type) cells had descending axon collaterals (13) with a consistent stellate-like dendrite morphology and high sag. The two cell types with no matching t-types within mouse L1, SST BAGE2, and VIP PCDH20 showed extremely diverse dendritic and axonal structure, often with substantial horizontal or descending axon branches—even avoiding L1 entirely in the case of some BAGE2 cells. These t-types appeared more uniform electrophysiologically, with relatively small spikes and high adaptation and sag, but were sparsely sampled and thus difficult to fully characterize.

In a few instances, we also observed differentiation between t-types within the same subclass. Within the LAMP5 subclass, sag and initial firing rate decreased from the NMBR t-type to DBP to LCP2 t-types (Fig. 2C, strip plots within boxplot; Spearman correlation FDR-corrected, $P < 10^{-3}$). Whereas other LAMP5 t-types were mostly restricted to L1 and superficial L2, the LCP2 t-type was found distributed across all cortical layers, with axonal arbors becoming less elongated and overlapping less with dendritic arbors for deeper cells (fig. S2, B and D, and fig. S3B). PAX6 cells were distinguished by whether the initial high-frequency firing formed a discrete burst (TNFAIP8L3) or continuously adapted (CDH12), and the two MC4R t-types were distinguished by the magnitude of sag and irregularity of firing (Fig. 2B).

Given the potential for the observed neuronal diversity to be determined in part by diversity in tissue-donor characteristics, we tested all morphoelectric features for effects of donor medical condition, sex, and age (fig. S6, A and B, and data S3). Most effects were small and in features not linked to L1 diversity, with the exception of higher dendritic branching in cells from tumor patients compared with that found in epilepsy patients (fig. S6A); this result was not explained by brain area or subclass.

Cross-species differences in L1

Evolutionary expansion of L2 and 3 (L2/3) in primates was previously linked to changes in cytoarchitecture—including thinning out of cell density and increased soma size—accompanying the specialization of pyramidal cell types (29). Therefore, we first quantified cytoarchitecture differences in L1 of human tissue samples [neuronal nuclei (NeuN)-stained slices from Patch-seq tissue blocks] compared with L1 in mouse samples (Fig. 3A). In contrast to cross-species differences in L2/3, human L1 was thicker but not less dense, with smaller cell

bodies as compared with several mouse neocortical areas (fig. S6D).

To study cross-species differences in the morphoelectric properties of L1 neurons, we compiled a comparison Patch-seq dataset from mouse L1 neurons ($n = 272$, 255 with electrophysiology and 43 with morphology) consisting of previously published data from a cross-layer analysis of interneurons in VISp (24) and additional recordings in L1 and L2/3 of visual cortex and the temporal association area (TEa). [TEa was held out as a validation set because this region is often compared with human MTG (38, 39).] Despite the differences in L1 cytoarchitecture, morphologies of L1 neurons generally showed remarkable similarity across species when comparing across matched homology-driven subclasses, with two exceptions (Fig. 3C, figs. S4 and S5, and data S4). No mouse L1 neurons had morphologies resembling the unmatched human L1 t-types (VIP PCDH20 and SST BAGE12, homologous to deeper mouse t-types), and the MC4R subclass was less morphologically distinct, providing evidence for type-specific divergence across species (fig. S5). Although human neurons across subclasses were slightly larger in horizontal extent, no differences were observed in vertical dimensions or dendritic diameter (fig. S3C). Mouse VIP cells had descending axon branches as in human VIP cells, but with greater variability of structure. Mouse LAMP5 cells had dense NGFC-like axonal arbors, confirming their match through *Ndnf/Npy* expression. Unlike human axons, mouse axons rarely extended above dendrites (Fig. 3D, left), perhaps reflecting sublaminal structure found only in the thicker human L1. Human neurites were also structured differently, with smaller contraction ratios (higher tortuosity) compared with the straighter but more heavily branched mouse dendrites (Fig. 3D and data S4).

As in previous studies, electrophysiological properties showed stronger differences across species (40). Mouse cells had less sag, lower amplitude spikes, and higher rheobase across types and within at least two of four matched subclasses (Fig. 3D and data S4). We replicated these findings in a comparison between MTG and a smaller TEa dataset, verifying that cross-species differences were not due to regional differences between MTG and VISp (Fig. 3E). Proportions of L1 t-types also varied little across brain regions in mouse and human single-neuron and -cell RNA-seq reference datasets (fig. S2B) (41). We also explored dependence of L1 interneuron morphoelectric properties on brain region within our human data and found moderate effects on a set of features including dendrite extent and input resistance (fig. S6C). For these features, more sparsely sampled regions either failed to differ from MTG cells or had smaller dendritic extent and larger input resistance, diverging more strongly from mouse

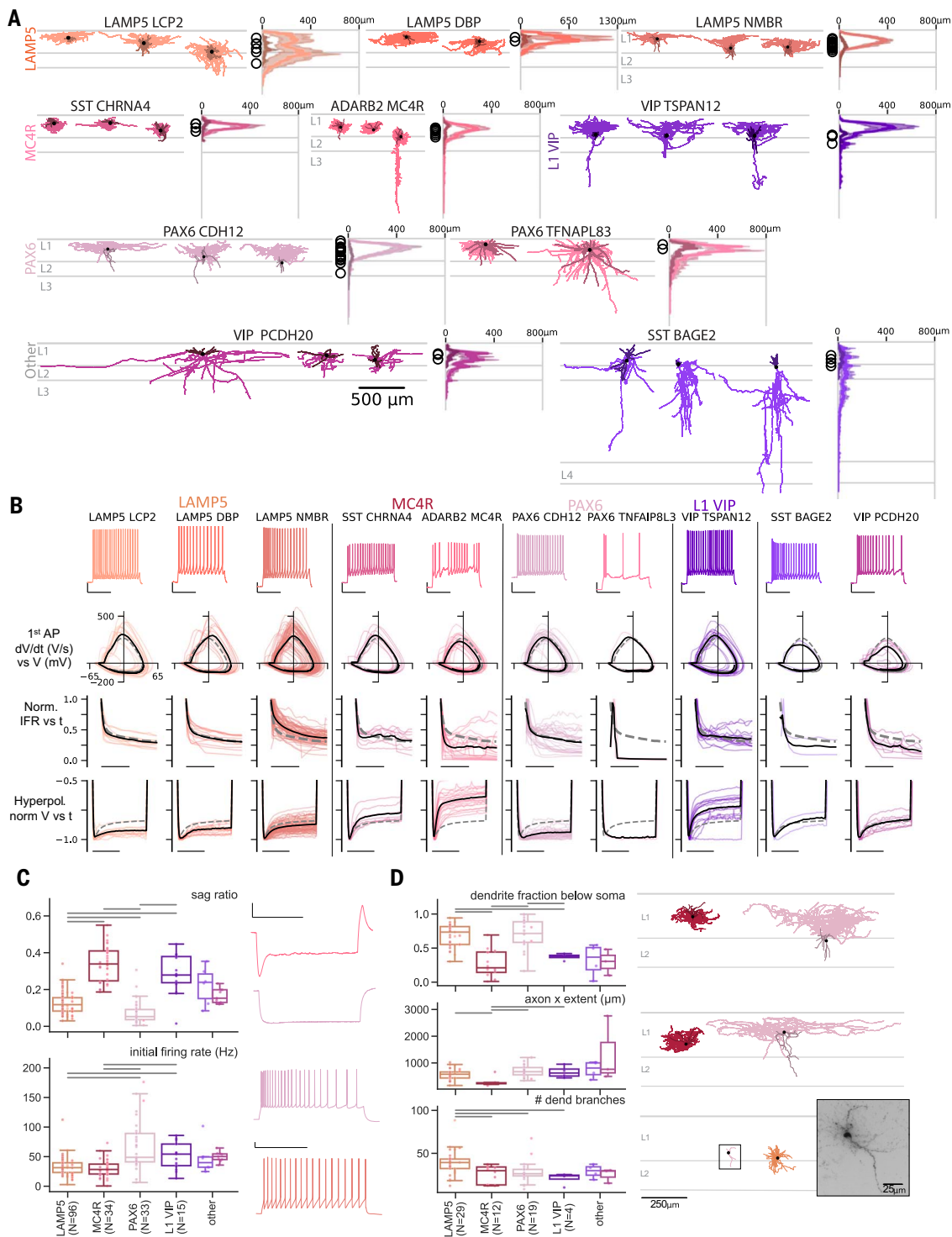


Fig. 2. Human L1 transcriptomic subclasses are morphoelectrically diverse.

(A) Example human morphologies for L1 t-types are displayed by subclass. Neurons are shown aligned to an average cortical template, with histograms to the right of the morphologies displaying average dendrite (darker color) and axon (lighter color) branch length by cortical depth for all reconstructed cells in L1 and L2 (shading shows ± 1 SD about mean; soma locations are represented by black circles). (B) Electrophysiological summary view by t-type and subclass. (Top) Images show example spiking responses. Scale bar, 0.5 s, 20 mV. (Bottom)

Cell-by-cell summary traces are shown, with t-type average (curved black lines), dataset average (curved dashed lines), and individual cells (curved colored lines). (Top to bottom) Diagrams show the phase-plane (dV/dt versus V) plot of first action potential. The instantaneous firing rate (IFR) and the hyperpolarizing response (Hyperpol.) are normalized to peak. Spiking plots (example, phase-plane, IFR) are at 40 pA above rheobase; hyperpolarizing response is at membrane potential closest to -100 mV. Scale bar, 0.5 s, 20 mV. Counts are found in table S2. AP, action potential. (C and D) Electrophysiological

and morphological features distinguishing L1 subclasses [Kruskal-Wallis (KW) test, FDR-corrected $P < 10^{-7}$ for electrophysiology, $<10^{-3}$ for morphology; data S2]. Boxplots show subclass statistics [box marks quartiles; whiskers extend $1.5 \times$ interquartile range (IQR) past box], with individual cells arranged horizontally by t-type. Significant pairwise comparisons marked by

lines above (FDR-corrected $P < 0.05$, Dunn's post hoc test). Illustrative electrophysiology traces (scale bar, 0.5 s, 20 mV) or layer-aligned morphologies are shown for high and low values of each feature. (Inset) Image shows that sparsity of dendrites in human PAX6 cells is not due to inability to resolve dendrites.

features, which suggests again that the cross-species differences were not inflated by regional sampling.

We next investigated causal factors underlying cross-species electrophysiology differences. Simulations of passive biophysical models based on reconstructed morphologies showed that input-resistance variability in human but not mouse L1 cells can be explained by morphology, suggesting that the lower input resistance in mouse may be partly due to active ionic conductances (fig. S3G). The small cross-species effect of morphology on input resistance in the models could be explained by differences in dendritic branching, especially near the soma, which would affect the effective membrane area for leak conductance. Indeed, we found a higher peak of total dendrite cross-sectional area at $\sim 50 \mu\text{m}$ from the soma—as well as slightly higher total volume in mouse cells (fig. S3F)—supporting this explanation. We also looked for correlated differences in ion-channel gene expression and morphological features compared with membrane properties in the Patch-seq dataset. Differences in spike shape and threshold could be explained by potassium-channel differences, along with related features such as rheobase and delayed spiking. Indeed, the expression of genes (*KCND2*, *KCND3*, and *KCNH7*) associated with fast-inactivating (A-type) voltage-gated K^+ (Kv) channels [Kv4.2, Kv4.3, and the ERG3 channel Kv11.3 (42, 43)] was higher in mouse neurons and was correlated with several action-potential features (fig. S3E). To test for corresponding differences in K^+ channel conductance, we measured macroscopic currents in nucleated patches following whole-cell recording in a subset of cells. Compared with human neurons, mouse neurons showed much higher A-type K^+ conductance but comparable slow-inactivating (D-type) conductance (Fig. 3F). Considering that blocking Kv4 channels in mouse NGFC cells decreases action-potential threshold and latency of first action-potential onset (44), these differences in A-type K^+ channel conductance, along with lack of Kv1.1 expression, may contribute to the lack of late spiking observed in human L1 NGFC cells as well (45).

Lastly, we asked whether the strong morphoelectric variability observed between human L1 subclasses is also present in L1 of mouse neocortex. Ranking electrophysiological and morphological features by the amount of variability between subclasses they explain, we found that the two species had a similar amount of variability (number of significantly different

features and their effect size) but varied along different sets of features (Fig. 3G). The most distinct features in human, such as sag and spike-shape adaptation, showed little variability in mouse, and unlike in human, mouse subclasses varied physiologically in interspike interval (ISI) adaptation and spike after-hyperpolarization (AHP) properties and morphologically in relative vertical positioning of the axonal arbor (most features are largely driven by L1 VIP subclass; figs. S3 and S5).

Distinctive neuronal phenotypes in human L1

Despite the quantitative similarity in L1 heterogeneity across species, we noted two particularly distinctive phenotypes found in human L1 only. The MC4R rosehip cells and the bursting PAX6 TNFAIP8L3 t-type were both qualitatively distinct from other human L1 types and did not appear similar to any mouse L1 types. To further highlight this contrast, we investigated each of these highly distinctive types in turn by quantifying the distinctive morphoelectric features and marker genes, then searching for comparable cells in the mouse L1 dataset.

Rosehip cells

The MC4R subclass, putative rosehip cells, comprises two transcriptomically similar t-types, SST CHRNA4 and ADARB2 MC4R, both highly distinct from other L1 types, including the LAMP5 LCP2 t-type originally identified with the rosehip phenotype (20). MC4R morphologies were all confirmed to qualitatively match the distinctive rosehip axonal structure and boutons (Fig. 4A and fig. S4) and were quantitatively distinct from other L1 types in terms of maximum axonal path distance and branch frequency (Fig. 4B). We also noted two examples of MC4R cells (both within the ADARB2 MC4R t-type) with elaborate descending axons reaching the lower half of L3 and confirmed that the characteristic large, dense axonal boutons were visible on both the central axonal arbor and descending axons when present (Fig. 4A, right). Electrophysiologically, both t-types that comprise the MC4R subclass showed strong sag, but only the ADARB2 MC4R t-type showed the distinctive irregular firing (45) and stronger and faster sag (Fig. 2B and Fig. 3, A and B). Cells in the ADARB2 MC4R t-type also had somas and axons localized near the L1/L2 border (fig. S2, C and D). We explored the expression of genes related to neuron physiology [ion channels and G protein-coupled receptors (GPCRs)] and found that markers distinguishing the entire rosehip subclass from the rest of L1,

including *HTR1F* [5-hydroxytryptamine (5-HT; serotonin) receptor 1F], along with markers distinguishing the rosehip subtypes—*GRM5* (metabotropic glutamate receptor 5) and *RELN* (reelin)—showed lower expression in ADARB2 MC4R neurons (Fig. 4C). Together, these differences in gene expression and physiology indicate that there are distinct rosehip neuron subtypes within human L1.

In mouse L1, we identified cell types corresponding to the MC4R subclass on the basis of similarity in transcriptomes, but this match was weak compared with the matches to other homologous types (Fig. 1D). Similarly, there were no mouse cell types observed with the morphological signatures consistent with human rosehip cells (fig. S5) and only partial matches to the electrophysiological signatures: The homologous mouse MC4R subclass had moderate sag but no irregular spiking (Fig. 4D). Irregular spiking resembling the human ADARB2 MC4R rosehip t-type was present only in a subset of LAMP5 cells (primarily Lamp5 Ntn1 Npy2r) that did not have other rosehip-like features (Fig. 4D). Although not directly matching the rosehip phenotype, the mouse homology-driven MC4R subclass carried similarities with the canopy cell (19). Matched characteristics included the moderate sag, along with gene expression (Fig. 1C, *Ndnf*⁺/*Npy*⁺) and wide dendritic extent, but mouse MC4R cells did not have the canopy's namesake L1a-dominant axon (fig. S7, B and C). Thus, the mouse MC4R subclass may be a distinct NGFC-like cell population but certainly lacks distinct boundaries that can be clearly resolved either by cross-species comparison or reference to previous mouse L1 classifications.

Bursting PAX6 TNFAIP8L3 cells

The other highly distinctive firing pattern in human L1 was in the PAX6 TNFAIP8L3 t-type, which fired in high-frequency bursts at the onset of stimulation, followed by quiescence or regular firing at higher stimulus amplitudes. Spiking and dendritic structure were qualitatively distinct between this t-type and the neighboring PAX6 CDH12 t-type, despite some similarity of axonal structure and subthreshold electrophysiology (Fig. 5A). Both the initial firing rate at rheobase and the after-depolarization potential (ADP) that followed the final spike quantitatively distinguished PAX6 TNFAIP8L3 cells from all other L1 cells (Fig. 5B), as did the number of dendritic branches and large horizontal dendritic extent, more than $550 \mu\text{m}$ wide.

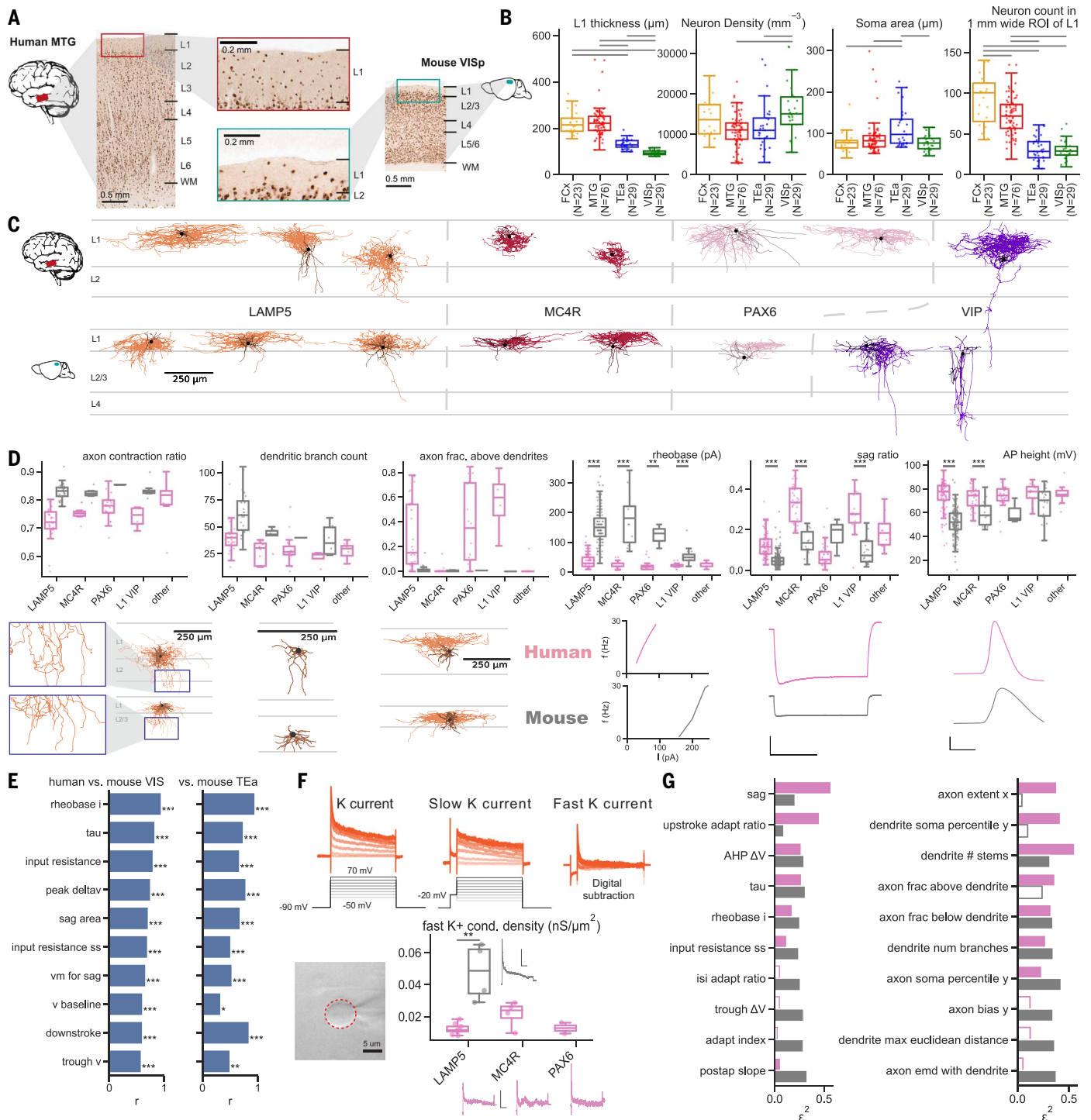


Fig. 3. Comparison of human and mouse L1. (A) Examples of NeuN labeling of neurons in human MTG and mouse VISp. (B) Comparisons of mouse versus human L1 thickness, neuron density, soma area, and neuron count in 1-mm wide regions of interest (ROIs) of L1. Metrics are plotted per ROI for L1 thickness, density, and neuron count, and per cell for soma area. Boxplots show quartiles, and asterisks indicate post hoc Dunn's test results (* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$) (calculated for MTG versus TEa only). Counts are found in Materials and methods. (C) Example layer-aligned morphologies from mouse and human L1 subclasses. One example is shown from each t-type; the scale

bar applies to both species. (D) (Left half) Morphological and (right half) electrophysiological features with differences between human and mouse L1 cells (species effect from two-way ANOVA on ranks, FDR-corrected $P < 10^{-38}$ for morphology, $P < 10^{-38}$ for electrophysiology). For features with a species-subclass interaction ($P < 0.05$, FDR-corrected for electrophysiology, not significant for morphology), asterisks indicate post hoc Dunn's test results (* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$). Counts are found in table S2. (Bottom) Representative examples from LAMP5 subclass are shown below each plot (Left to right: layer-aligned reconstructions; action potential frequency as a

function of current injection; response to hyperpolarizing current nearest -100 mV, scale bar 0.5 s, 10 mV; first action potential at rheobase, scale bar 1 ms, 20 mV). (E) Electrophysiological feature differences between human L1 and mouse VISp L1 (left) were validated by testing against mouse TEa (right). Features are selected by largest effect size against TEa [Mann-Whitney (MW) correlation coefficient (r), rank-biserial correlation]. Asterisks indicate significance (FDR-corrected MW test, $*P < 0.05$; $**P < 0.01$; $***P < 0.001$). (F) Nucleated patch recordings quantifying A-type K^+ conductance of L1 neurons.

Example traces show voltage commands (black) and recorded currents (orange) from measurement protocol (top), along with example soma size measurement. (Bottom) Boxplots show fast conductance density in both species, with example traces shown for each group (scale bars, 200 ms, 400 pA). Only LAMP5 neurons were sampled in mouse. (G) Features distinguishing L1 subclasses in human and mouse organized by relevance to each species. Bars show size of subclass effect (ϵ^2 from KW test), with features ranked by the difference between human and mouse effects. Unfilled bars indicate $P > 0.05$ (FDR-corrected).

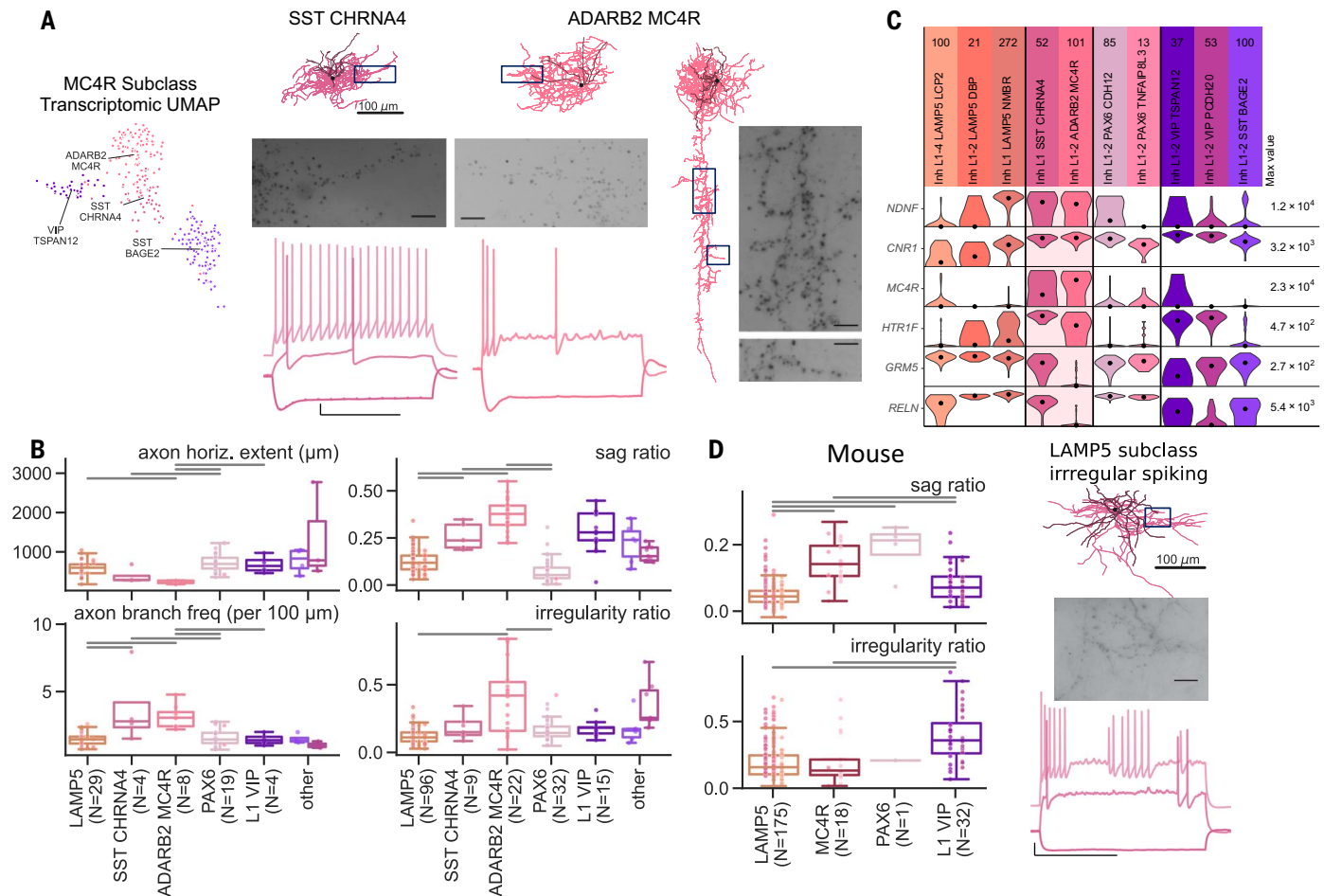


Fig. 4. MC4R rosehip cells. (A) Characterization of MC4R subtypes as rosehip cells. (Left) UMAP projection of transcriptomic data from MC4R and nearby subclasses. (Right) Example cells from each subtype. Morphologies show characteristic axonal arbors and boutons (Insets) 63x maximum intensity projection (MIP) images, scale bars, $10 \mu\text{m}$; compare to (D). Electrophysiology traces show sag response (hyperpolarization near -100 mV, rheobase, and rheobase $+40$ pA if present; scale bar, 0.5 s, 10 mV). (B) Electrophysiological and morphological features distinguishing MC4R t-types (FDR-corrected MW test versus rest of L1, $P < 10^{-4}$). Boxplots show statistics of MC4R subtypes and other subclasses (box marks quartiles; whiskers extend $1.5 \times \text{IQR}$ past box). Significant

pairwise comparisons (to MC4R t-types only) marked by lines above (FDR-corrected $P < 0.05$, Dunn's test post hoc to KW test). (C) Gene expression of MC4R subclass (highlighted) and other L1 t-types, for between- and within-subclass marker genes (snRNA-seq). Violin plots show expression in $\log(\text{CPM}+1)$, normalized by gene (maximal expression noted at right). (D) Characterization of mouse L1 cells with moderate sag or irregular firing, the human rosehip type's distinct properties [boxplots as shown in (B), all pairwise comparisons tested]. Example morphology and electrophysiology are shown for mouse Lamp5 Ntn1 Npy2r cell with highly irregular firing but lack of rosehip-like morphology [electrophysiology traces as shown in (A)]. (Inset) Axonal boutons, scale bar, $10 \mu\text{m}$.

In mouse L1, the homologous PAX6 subclass was extremely rare, comprising only a few cells in the Lamp5 Krt73 t-type. These cells tended to fire in doublets at stimulus onset rather than in a full burst, sometimes followed by a delayed ADP (Fig. 5C). Some cells in the Lamp5 Fam19a1

Pax6 t-type also showed this firing pattern (fig. S7D), likely the same subset that aligns transcriptomically to the human PAX6 subclass (Fig. 1C). Mouse doublet-firing cells also showed a depolarizing "hump" for current injection just below rheobase (Fig. 5D), which together with the

marker-gene signature (*Ndnf*/*Vip*⁻/*Chrna7*⁺) identifies them as mouse $\alpha 7$ cells, a type previously defined in mouse by these physiological and gene features (19). This hump was suggested to indicate activation of T-type calcium channels, likely the same mechanism underlying

the bursting in human cells (45). Bursting was also previously noted in a subset of mouse single-bouquet cells (SBCs) (25), a group defined by loose morphological criteria. This class likely overlaps with the doublet-firing t-types (46), suggesting that they may burst under different physiological conditions.

Using these insights from the cross-species alignment, we explored the expression of related genes in the human PAX6 t-types (Fig. 5C). Both t-types matched the $\alpha 7$ marker gene signature (along with the SST BAGE2 t-type; *Ndnf/Vip⁻/Chrna7⁺*) and strongly expressed the T-type calcium channel α subunit gene *CACNA1G*, highlighting T-type calcium channels as a potential factor in the burst and doublet firing across species (fig. S7A). Given the lack

of bursting in PAX6 CDH12 cells, other ion-channel genes differentially expressed between the two human PAX6 t-types likely also play a role, including *TRPC3*, a nonspecific cation channel that can regulate resting membrane potential (47).

Cross-modality relationships of L1 subclasses and t-types

Given the multiple observations of distinctness between human types in contrast with continuous variation between mouse types, we explored this difference more comprehensively by defining a common quantitative framework for distinctness across modalities. We generalized the d' metric used for transcriptomic distinctness (23, 24), quantifying

the performance of classifiers trained to distinguish pairs of t-types on the basis of electrophysiological and morphological features. The resulting t-type similarity matrices (Fig. 6A) showed comparable subclass structure in both electrophysiology and transcriptomics, with smaller d' values within subclass blocks and higher values outside. Of note is that d' metrics were highly correlated between modalities, demonstrating that cell types with distinctive transcriptomes have similarly distinctive electrophysiological properties (Pearson's $r = 0.59$, $P = 0.00016$) (Fig. 6B). The single within-subclass pair with a high d' was LAMP5 NMBR and LAMP5 LCP2, which sit at opposite ends of the LAMP5 continuum. We also calculated d' similarity matrices at the subclass level to

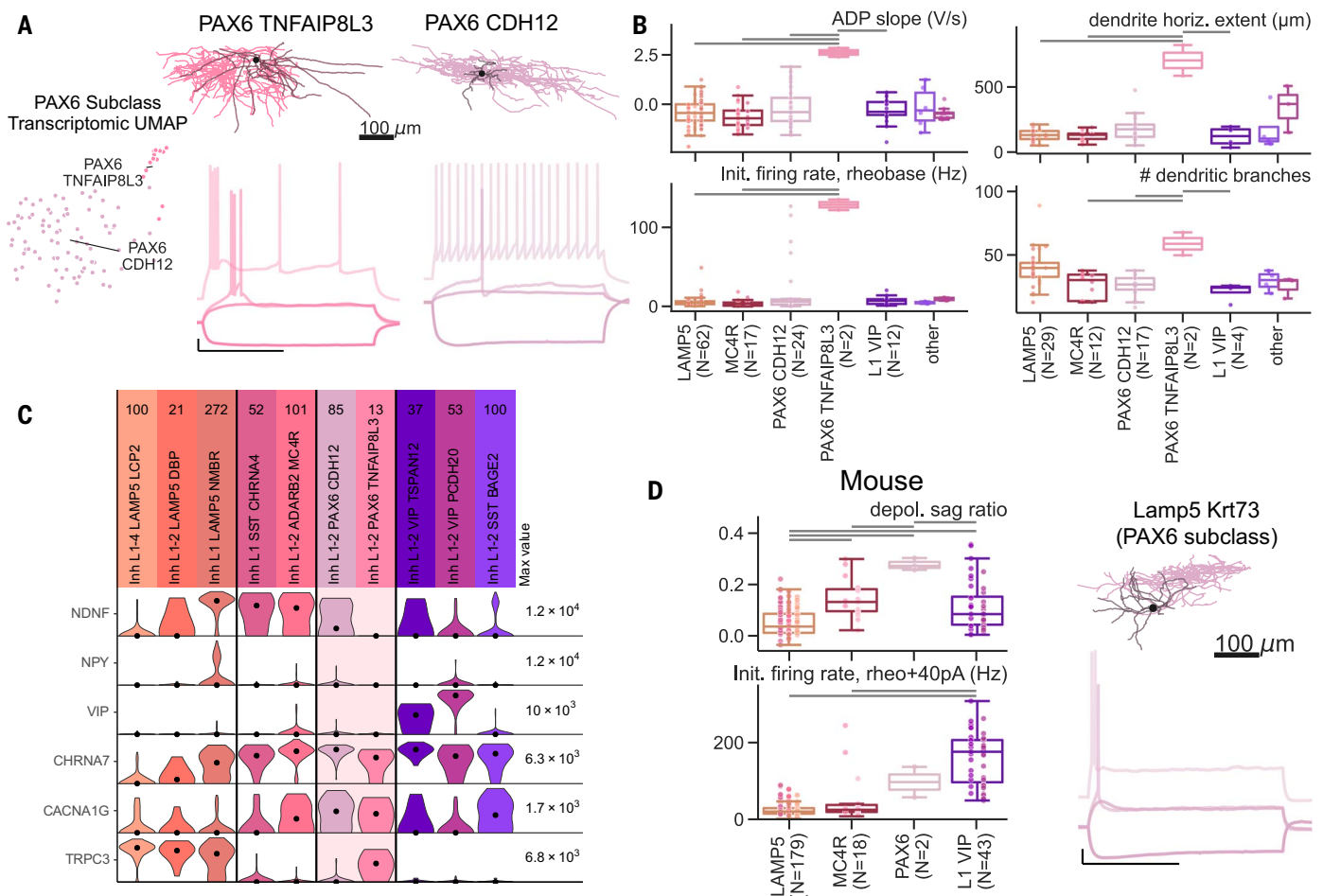


Fig. 5. Burst spiking PAX6 TNFAIP8L3 cells. (A) Reconstructed morphologies and example electrophysiology for bursting and nonbursting PAX6 t-types (TNFAIP8L3 and CDH12) (hyperpolarization near -100 mV, depolarization below rheobase, spiking at rheobase and rheobase +40 pA; scale bar, 0.5 s/10 mV). (Inset) UMAP projection of transcriptomic data from PAX6 subclass is shown. (B) Electrophysiological and morphological features distinguishing PAX6 TNFAIP8L3 t-type (FDR-corrected MW test versus rest of L1, $P < 0.05$ for morphology, $P < 0.01$ for electrophysiology). Boxplots show statistics of PAX6 subtypes and other subclasses (box marks quartiles; whiskers extend $1.5 \times \text{IQR}$ past box). Significant pairwise comparisons (to PAX6 TNFAIP8L3 only) are marked by

horizontal lines above (FDR-corrected $P < 0.05$, Dunn's test post hoc to KW test). (C) Gene expression of human PAX6 subclass (highlighted) and other L1 t-types, for $\alpha 7$ type and bursting-related marker genes (snRNA-seq). Violin plots show expression in $\log(\text{CPM}+1)$, normalized by gene (maximal expression noted at right). (D) (Left) Characterization of mouse L1 cells with initial doublet firing in terms of the human PAX6 TNFAIP8L3 type's distinct properties. (Right) Example morphology and electrophysiology shown from PAX6 subclass (Lamp5 Krt73 t-type). Depolarizing sag ratio is the normalized size of the hump at stimulus onset just below rheobase [electrophysiology traces as in (A); boxplots as in (B), all pairwise comparisons tested].

allow comparison between species in all three modalities (Fig. 6C). These results confirmed the generally higher transcriptomic distinctness of subclasses in human and show that in mouse, the VIP subclass was highly distinct in all modalities, with other subclasses generally less distinct.

To visualize the subclass-level distinctness in terms of specific electrophysiological features, we found the pair of features that most distinguished each subclass and showed that clusters defined by these features closely match

the transcriptomic subclass boundaries (Fig. 6D). We also tested the effectiveness of assigning subclass labels to neurons based on the full electrophysiological feature set. A multiclass classifier evaluated by cross-validation on the primary dataset had 82% accuracy balanced across subclasses (fig. S8A). To mimic the out-of-sample issues that could be encountered for future L1 datasets collected under different conditions, we also tested classifier performance on data held out of our primary analysis because of equipment and protocol differences.

After excluding features for which the distributions strongly differed from the primary dataset, we found comparable classification performance (81%, fig. S8B), reinforcing the utility of the human L1 subclasses for understanding L1 variability even in the absence of transcriptomic information to assign subclass identity.

Discussion

Using Patch-seq, we identified a coherent view of human L1 interneurons in which neuronal

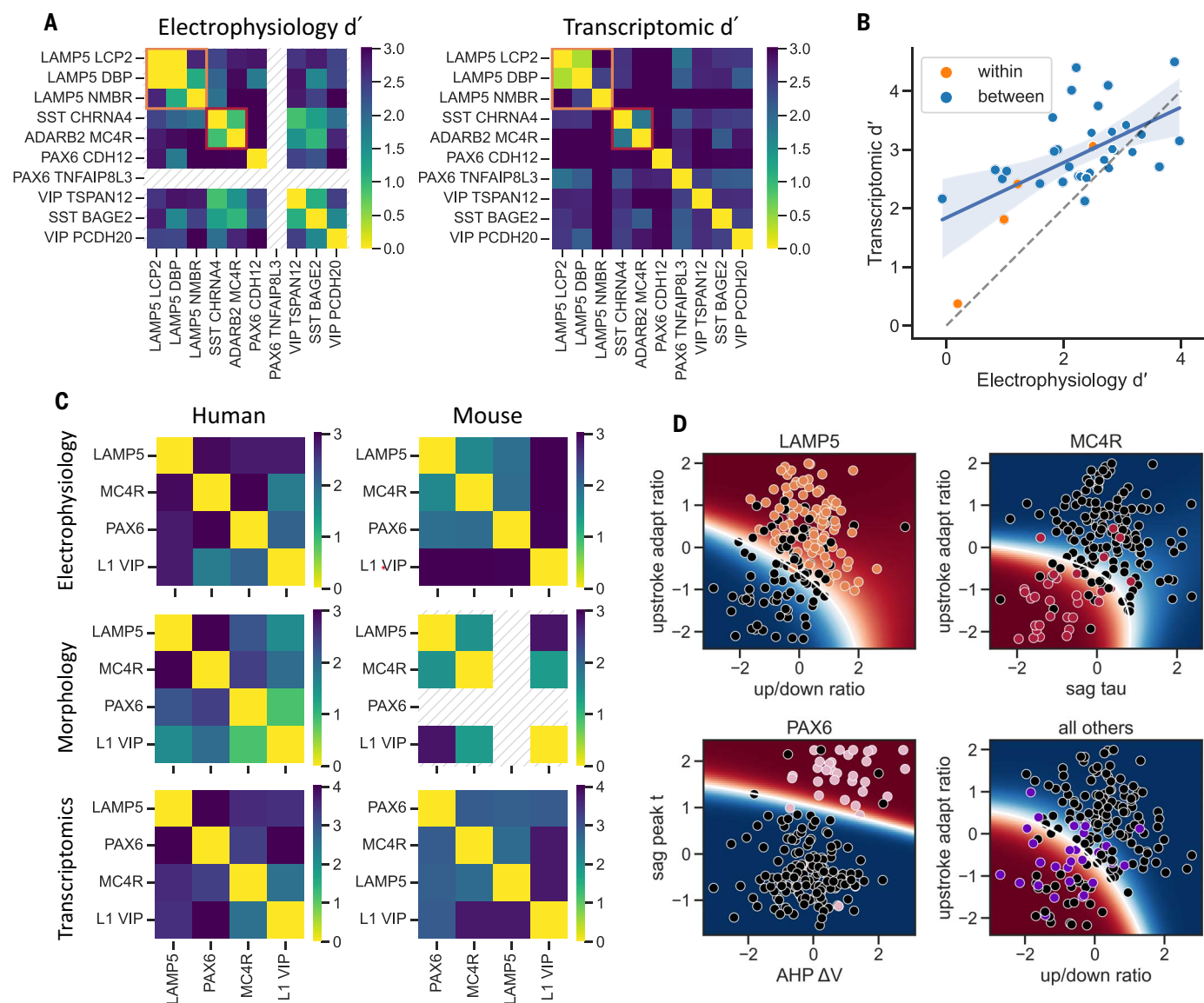


Fig. 6. Quantifying distinctness of L1 t-types and cross-modality structure.

(A) Pairwise distinctness of human L1 t-types, from classifiers using electrophysiology (left) and gene expression (right). d' (d prime) is a metric of separation of distribution means, scaled relative to the SD. Groups with $N < 4$ are excluded (hatched area). (B) Correlation of pairwise d' values between transcriptomic and electrophysiological feature spaces. Pearson's $r = 0.59$; $P = 0.00016$; shading shows bootstrapped 95% confidence interval of regression. Within-subclass pairs are shown in orange to confirm subclass structure.

(C) Pairwise distinctness d' of L1 subclasses across species and data modality.

Groups with $N < 10$ are excluded. (D) Clustering of human L1 cells in electrophysiology subspaces, with correspondence to L1 subclasses. Points show all L1 neurons, with the subclass of interest in color. Background color shows cluster membership likelihoods from two-cluster Gaussian mixture model trained on unlabeled data. F1 scores: LAMP5, 0.81; MC4R, 0.69; PAX6, 0.89; all others, 0.5 (L1 VIP and ungrouped t-types). All features are normalized and Yeo-Johnson transformed to approximate Gaussian distribution.

subclasses defined by transcriptomic distinctness have similarly distinct morphoelectric phenotypes. Although mouse L1 neurons had a similar range of diversity in most features, the features that distinguished cell types were different between species, and human L1 neurons spanned a wider range of sizes. In addition to cell types not found in mouse L1 (VIP PCDH20 and SST BAGE2), two human cell types emerged with especially distinct phenotypes that were not matched in their putative homologs in mouse: the compact, high-sag MC4R rosehip subclass and the large, burst-spiking PAX6 TNFAIP8L3 t-type. Human and mouse neurons also showed consistent differences in certain morphological and physiological properties across all subclasses, despite a general similarity in cell size. These results indicate a general conservation of L1 inhibitory neuron diversity but with distinct specializations in cell properties and subclass and cell-type proportions, likely leading to differences in the regulation of higher-order input to the human cortical circuit.

Categorizing L1 neuron types

We provide support for previous classification schemes in mouse consisting of four primary types in L1 but also clarify a need for precise data-driven criteria for those types. Homology-driven subclasses were nearly aligned with cell-type classifications based on single marker genes (19), but this alignment was often ambiguous and lacked coherence across modalities. Similarly, other coarse single-modality cell-type distinctions in L1 (late spiking versus non late spiking, NGFC versus SBC) likely grouped multiple distinct subclasses and shifted the exact boundaries (48). Our results suggest that a larger role for continuous variability should be considered when studying cell-type diversity in mouse L1. A continuous transition between *Ndnf*⁺/*Npy*⁺ NGFC cells and *Ndnf*⁺/*Npy*⁺ canopy cells was previously noted (24), corresponding to similarity between the Lamp5 Ntn1 *Npy*2r t-type (LAMP subclass) and Lamp5 Fam19a1 Tmem182 (MC4R subclass). We also observed continuity between mouse PAX6 (partial α 7 cell match) and MC4R (partial canopy-cell match) types, with α 7-like doublet spiking in some MC4R cells. Additionally complicating the view of canopy cells, the best-match cells (MC4R subclass) had some properties at odds with the original definition, expressing *Chrna7* and missing the canopy-like L1a axons, which were observed primarily in *Npy*⁺ LAMP5 cells (fig. S7, A to C). These ambiguous subclass boundaries are a strong contrast to the clear cross-modality subclass distinctions in human L1, perhaps related to the smaller number of well-resolved transcriptomic types in mouse.

We also demonstrate the benefits of detailed transcriptomic data over small sets of genetic markers for both accurately characterizing cell-

type divisions and establishing cross-species homologies that facilitate comparative analysis. Within species, reliance on marker genes can overstate the distinctness of cell types, as with the NGFC/canopy distinction, or even lead to misidentification, as with the original description of human L1 rosehip cells, which were assigned an incorrect t-type on the basis of observed marker-gene patterns (20). Across species, the lack of conserved L1 markers was striking and perhaps exclusive to L1. Even using the full transcriptome, previous work on this homology found variable results for L1 types with different brain areas and methods (21). In this work, we chose to quantify similarity of t-types across species in a way that better captured ambiguity, finding strong matches for some types and weaker for others. In general, all transcriptomic homology matches were supported or contradicted by morphoelectric comparisons. Ambiguous matches may indicate areas of evolutionary change that could be illuminated by comparative or developmental analyses of additional species phylogenetically related to mouse and human. For instance, the weak cross-species transcriptomic similarity of the MC4R subclass (Fig. 1C) and lack of phenotypic similarity together suggest that human and mouse MC4R cells could represent distinct innovations in each taxon, rather than a true homology.

In human L1, the strong alignment of subclass distinctions across modalities suggests that cells can be classified by using only morphology or electrophysiology with reasonable accuracy. Condition-dependent variation of certain electrophysiological features can present a challenge to this approach, however, especially for classifications that rely on small numbers of features with especially strong qualitative variability. We failed to observe two electrophysiological phenotypes that had been noted in past work: late spiking in human NGFCs (40) and full bursting in mouse SBCs (25), qualitative features for which conflicting observations have also been reported (19, 49). Although potential contributing factors are numerous (age differences; donor characteristics; and recording conditions, including internal solutions, temperature, and equipment differences), we showed that for classification with a large feature set, identifying and excluding affected features can rescue reliable performance.

Cell types, evolution, and function

We propose that the divergence in the L1 interneuron repertoire between mouse and human partly reflects the increasingly complex role of L2/3 in the primate neocortical circuit (29, 50, 51); new types of pyramidal cells might have necessitated new types of dendritic inhibition. Compared with mouse, human L2/3 excitatory neurons are more transcriptomi-

cally distinct and show larger sublamina distinctions in gene expression, dendritic morphology, and physiology. As in L1, there are also neuronal types in L2/3 of human MTG with no clear homolog in L2/3 of mouse neocortex (29). Although these observations suggest similar degrees of evolutionary divergence in L2/3 excitatory neurons and L1 inhibitory interneurons, the L1 circuit might also have adapted to changes in deep-layer pyramidal-neuron populations, including the decreased proportions in primates of L5 extratelencephalic pyramidal neurons, the prominent apical dendrites of which are targeted by L1 inhibition in rodents (21, 22, 52).

Two types of interneurons stood out as especially distinct within human L1, with the potential to contribute to functional differences between human and mouse: the MC4R rosehip cells and the PAX6 TNFAIP8L3 t-type. Rosehip cells were previously shown to inhibit pyramidal-cell apical-dendrite shafts in L2/3 (20); the rosehip subtypes, with distinct electrophysiology, could plausibly perform similar but distinct inhibitory functions or selectively modulate different pyramidal-neuron subtypes in L2/3 (29). For example, the irregular firing of the MC4R rosehip t-type suggests that this cell type is modulated by input in the beta frequency band (20), whereas the regular firing of the CHRNA4 rosehip t-type suggests a lack of beta-band modulation. The strong bursting dynamics and distinctive morphology of the PAX6 TNFAIP8L3 t-type also clearly point to a distinct functional role compared with that of neurons in neighboring subclasses or even the more closely related PAX6 CDH12 t-type. Their extended dendrites are well positioned to integrate local pyramidal-cell inputs across a broad spatial footprint and long-range axonal inputs across topographic boundaries, and the bursting would provide a strong immediate activation in response to strong or coincidental input. The clear identification of a cross-species homology for the PAX6 subclass can help in deciphering its function, combining functional insights gained from manipulating mouse cells with indirect insights from the more distinctive morphologies of human cells. Conflicting connectivity patterns have been observed for coarser cell types that likely include some PAX6 cells: Mouse α 7 cells synapse onto nearby L2 pyramidal neurons (8), whereas rat SBCs synapse onto L2/3 interneurons (53). Thus, more focused investigation of PAX6 cell connectivity is needed to illuminate the function of this subclass, and in turn, the functional implications of specialization within this subclass in human L1.

In addition to these distinct types that we identified in human, we observed several subclass-independent cross-species differences in morphoelectric properties between species, which likely have functional consequences.

Forms of morphoelectric variability found in human and not mouse L1 include sag and spike adaptation properties. These vary both between and within subclasses in human and may contribute to differences in the spectral selectivity of the L1 circuit. For example, the higher-voltage sag response in human L1 (and variation between LAMP5 subtypes) may contribute to differences in the temporal summation of synaptic input, as in L2/3 pyramidal neurons (54, 55). The separation of dendritic and axonal arbors in human L1 cells, which is not found in mouse, indicates the possibility of more sublaminal structure in L1 microcircuits in human. Similarly, given the sublaminal selectivity of thalamic projections to *Vip*⁺ versus *Ndnf*⁺ cells in mouse L1 (56), the human L1 VIP types, which are located in deeper layers in mouse, may be especially relevant to thalamocortical microcircuit structure. The higher input resistance and lower rheobase in human imply increased sensitivity to low-intensity synaptic input, which could have direct computational functions or help compensate for circuit differences, such as changes in excitatory to inhibitory cell ratios between mouse and human (21, 22, 32). In particular, these cross-species morphoelectric differences may indicate varying demands on the NGFC subclass, which was otherwise largely conserved across species. NGFC GABAergic (both synaptic and extrasynaptic) and gap junction-mediated transmission depend critically on the spatial properties of the axonal arbor (37, 57, 58), but the increase in arbor size in human LAMP5 cells (~1.2×) was much less than the 1.6× increase in pyramidal-cell apical dendrite extent in L1. Changes in input sensitivity could enable the conservation of their “blanket” inhibitory function while also permitting some increased spatial and topographic selectivity; NGFC circuit connectivity has been shown to be both tightly controlled (59, 60) and to exert strong effects on pyramidal-cell sensory processing (1, 6).

Much experimental evidence has documented the importance of neuromodulatory control on L1 function (40, 61–63). Linking distinct subclasses in L1 to detailed gene expression data (including emerging spatial transcriptomics results) can lead to refined hypotheses for cell-type specific neuromodulation. In particular, human MC4R cells distinctly expressed several modulatory receptor genes (Fig. S9A), including the melanocortin receptor *MC4R*, which plays a role in energy homeostasis in hypothalamus (64), serotonin receptor *HTR1F*, and metabotropic glutamate receptor *GRM1* (along with differential expression of *GRM5* between MC4R t-types). Cholinergic activation of L1 cells (61), suggested to control attention, may also differentially modulate MC4R cells relative to other L1 types as a result of their stronger *CHRNA6* and *CHRNA4* expression.

Our findings have a few limitations related to studying human cell types with the Patch-seq

approach. Certain cell types were undersampled, including those with no mouse homolog in L1 (VIP PCDH20 and SST BAGE2), limiting our ability to fully characterize their properties. Future work using viral genetic labeling approaches to label difficult-to-target types could help in this regard (65). Our cross-species comparisons were limited to differences between mouse and human and thus could not elucidate evolutionary progression of the L1 interneuron repertoire or truly reveal human specializations. Additional investigations of L1 cell types in other species would help, whether based on single-cell transcriptomics alone or full characterization through Patch-seq. Even in mouse, our analysis focused on comparisons to human and thus did not fully resolve different views of mouse L1 subclasses. A definitive answer may require a more comprehensive Patch-seq dataset, analyzed from a cross-species perspective and perhaps in concert with multi-region snRNA-seq and spatial transcriptomics (66). Lastly, directly relating any differences in intrinsic physiology, morphology, or gene expression to functional consequences requires further investigation, including characterizing synaptic connectivity and neuromodulation.

Nonetheless, the comprehensive human L1 Patch-seq dataset reported here has generated a wealth of hypotheses to inspire future work, along with tools to support that work. Our analysis provides tools for the classification of L1 diversity in both human and mouse, insights into functional relationships underlying physiological differences, and the clear identification of subclasses and subtypes that are likely to be of particular interest in functional studies—all important steps for deciphering the function of this enigmatic layer of neocortex. These data may also offer insight into various disease states, considering the importance of inhibition in circuit dysfunction (67, 68). This approach represents a roadmap for annotating functionally related properties onto transcriptomically defined cell-type taxonomies that could be applied across the primate brain—a crucial step toward linking cell-type diversity to functional diversity within a neural circuit.

Materials and methods

Detailed descriptions of Patch-seq data collection methods in the form of technical white papers can also be found under “Documentation” at <http://celltypes.brain-map.org>.

Human tissue acquisition

Surgical specimens were obtained from local hospitals (Seattle - Harborview Medical Center; Swedish Medical Center and University of Washington Medical Center; Amsterdam - Vrije Universiteit Medical Center; and Szeged - Department of Neurosurgery, University of Szeged) in collaboration with local neurosurgeons. Data included in this study were

obtained from neurosurgical tissue resections for the treatment of refractory temporal lobe epilepsy, hydrocephalus, or deep-brain tumor (data S1). All patients provided informed consent, and experimental procedures were approved by hospital institute review boards before commencing the study (IRBs 1111798 and 49119). Tissue was placed in slicing artificial cerebral spinal fluid (ACSF) as soon as possible following resection. Slicing ACSF comprised (in mM) 92 *N*-methyl-D-glucamine chloride (NMDG-Cl), 2.5 KCl, 1.2 NaH₂PO₄, 30 NaHCO₃, 20 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), 25 D-glucose, 2 thiourea, 5 sodium-L-ascorbate, 3 sodium pyruvate, 0.5 CaCl₂·4H₂O, and 10 MgSO₄·7H₂O. Before use, the solution was equilibrated with 95% O₂ and 5% CO₂, and the pH was adjusted to 7.3 to 7.4 by addition of 5N HCl solution. Osmolality was verified to be between 295 and 310 mOsm kg⁻¹. Human surgical tissue specimens were immediately transported (10 to 35 min) from the hospital site to the laboratory for further processing.

Mouse breeding and husbandry

All procedures were carried out in accordance with the Institutional Animal Care and Use Committee at the Allen Institute for Brain Science or Vrije Universiteit. Animals (<5 mice per cage) were provided food and water ad libitum and were maintained on a regular 12-hour light:dark cycle; rooms were kept at 21.1°C and 45 to 70% humidity. Mice were maintained on the C57BL/6J background (RRID:IMSR_JAX:000664), and newly received or generated transgenic lines were backcrossed to C57BL/6J. Experimental animals were heterozygous for the recombinase transgenes and the reporter transgenes. For details on transgenic lines, age, or other details see data S1.

Tissue processing

Data were obtained from male and female mice between the ages of postnatal day (P)45 and P70. Mice were anaesthetized with 5% isoflurane and intracardially perfused with 25 or 50 ml of 0° to 4°C slicing ACSF. Human or mouse acute brain slices (350 μm) were prepared with a Compresstome VF-300 (Precisionary Instruments) or VT1200S (Leica Biosystems) vibrating blade microtome modified for block-face image acquisition (Mako G125B PoE camera with custom integrated software) before each section to aid in registration to the common reference atlas.

Slices were transferred to a carbogenated (95% O₂/5% CO₂) and warmed (34°C) slicing ACSF for 10 min, then transferred to room temperature holding ACSF of the composition (72) (in mM): 92 NaCl, 2.5 KCl, 1.2 NaH₂PO₄, 30 NaHCO₃, 20 HEPES, 25 D-glucose, 2 thiourea, 5 sodium-L-ascorbate, 3 sodium pyruvate, 2 CaCl₂·4H₂O and 2 MgSO₄·7H₂O

for the remainder of the day until transferred for patch-clamp recordings. Before use, the solution was equilibrated with 95% O₂ and 5% CO₂, and the pH was adjusted to 7.3 using NaOH. Osmolality was verified to be between 295 and 310 mOsm kg⁻¹.

Patch-clamp recording

A step-by-step protocol for the Patch-seq methods used in this study can be found at <https://www.protocols.io/view/patch-seq-recording-and-extraction-detailed-protoc-5qpvozb2bg40/v2>. Slices were continuously perfused (2 ml/min) with fresh, warm (32 to 34°C) recording ACSF containing the following (in mM): 126 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 12.5 D-glucose, 2 CaCl₂·4H₂O, and 2 MgSO₄·7H₂O (pH 7.3), continuously bubbled with 95% O₂ and 5% CO₂. The bath solution contained blockers of fast glutamatergic (1 mM kynurenic acid) and GABAergic synaptic transmission (0.1 mM picrotoxin). Thick-walled borosilicate glass (Warner Instruments, G150F-3) electrodes were manufactured (Narishige PC-10 or Sutter Instruments P-87) with a resistance of 4 to 5 MΩ. Before recording, the electrodes were filled with ~1.0 to 2.0 μl of internal solution [110 mM potassium gluconate, 10.0 mM HEPES, 0.2 mM ethylene glycol-bis (2-aminoethylether)-N,N,N',N'-tetraacetic acid, 4 mM potassium chloride, 0.3 mM guanosine 5'-triphosphate sodium salt hydrate, 10 mM phosphocreatine disodium salt hydrate, 1 mM adenosine 5'-triphosphate magnesium salt, 20 μg ml⁻¹ glycogen, 0.5 U μl⁻¹ RNase inhibitor (Takara, 2313A), 0.02 Alexa 594 or 488, and 0.5% biocytin (Sigma B4261), pH 7.3]. The pipette was mounted on a Multiclamp 700B amplifier headstage (Molecular Devices) fixed to a micromanipulator (PatchStar, Scientifica or Mini25, Luigs and Neumann).

Electrophysiology signals were recorded using an ITC-18 Data Acquisition Interface (HEKA). Commands were generated, signals were processed, and amplifier metadata were acquired using MIES (<https://github.com/AllenInstitute/MIES/>), written in Igor Pro (Wavemetrics, RRID:SCR_000325). Data were filtered (Bessel) at 10 kHz and digitized at 50 kHz. Data were reported uncorrected for the measured -14-mV liquid-junction potential between the electrode and bath solutions.

Before data collection, all surfaces, equipment, and materials were thoroughly cleaned in the following manner: a wipe down with DNA away (Thermo Scientific), RNase Zap (Sigma-Aldrich) and finally with nuclease-free water.

L1 was identifiable in mouse and human brain slices as the neuron sparse region directly between the pial surface and the neuron-dense layer 2/3. Neurons within L1 were targeted for patch-clamp recordings.

After formation of a stable seal and break-in, the resting membrane potential of the neuron

was recorded (typically within the first minute). A bias current was injected, either manually or automatically using algorithms within the MIES data acquisition package, for the remainder of the experiment to maintain that initial resting membrane potential. Bias currents remained stable for a minimum of 1 s before each stimulus current injection. Upon attaining whole-cell current clamp mode, the pipette capacitance was compensated and the bridge was balanced.

The voltage response of each cell was recorded in response to a standardized stimulus paradigm described previously (26) that included square pulses, ramps, and chirps, with the goal of extracting features that could be compared across cells, rather than tailoring each stimulus to the physiological input of that neuron.

Nucleus extraction

Upon completion of electrophysiological examination, the pipette was centered on the soma or placed near the nucleus (if visible). A small amount of negative pressure was applied (~-30 mbar) to begin cytosol extraction and attract the nucleus to the tip of pipette. After approximately 1 min, the soma had visibly shrunk and/or the nucleus was near the tip of the pipette. While maintaining the negative pressure, the pipette was slowly retracted diagonally in the *x* and *z* direction. Slow, continuous movement was maintained while monitoring the pipette seal. Once the pipette seal reached >1 GΩ and the nucleus was visible on the tip of the pipette, the speed was increased to remove the pipette from the slice. The pipette containing internal solution, cytosol, and nucleus was removed from the pipette holder, and the contents were expelled into a PCR tube containing the lysis buffer (Takara, 634894).

Voltage clamp experiments

For a subset of experiments with a high nucleated patch resistance (>1000 MΩ), we measured macroscopic outward ionic currents in voltage clamp. To reduce capacitive artifacts, the pipette containing the nucleus was raised to the upper portion of the bath. For K⁺ channels, activation curves were constructed from 1-s depolarizing voltage commands (-50 to +70 mV in 10-mV voltage steps) from a holding potential of -90 mV. Linear leakage and capacitive currents were digitally subtracted by scaling traces at smaller command voltages in which no voltage-dependent current was activated. To isolate K⁺ currents into their phenomenological components (A-type, D-type, noninactivating), we exploited the voltage dependent properties of the putative channels underlying each component (73). The fast-inactivating component (*I*_{KA}) was inactivated by a brief step to -20 mV followed by a series of 1-s voltage steps ranging from -50 to 70 mV in 10-mV increments. *I*_{KA} was obtained by digi-

tally subtracting the resultant current from the total current, post hoc. Step depolarization to 70 mV from a holding potential of -20 mV was used to inactivate all K⁺ current and revealed a sustained current. Subtracting the sustained current from the current used to isolate *I*_{KA} revealed a slowly inactivating, D-type (*I*_{KD}) current. Peak currents were calculated for each voltage step. Conductance values were calculated based on the recorded membrane potentials and a K⁺ reversal potential at -100 mV. The surface area of the nucleated patch was calculated to obtain current and conductance densities.

Quality control

For an individual sweep to be included in analysis, the following criteria were applied: (1) membrane potential within 2 mV of target potential (initial resting potential of cell); (2) bias (leak) current 0 ± 100 pA; and (3) root mean square noise measurements in a short window (1.5 ms, to gauge high-frequency noise) and longer window (500 ms, to measure patch instability) <0.2 mV and 0.5 mV, respectively.

For human electrophysiology in the primary dataset, QC filters were also imposed at the cell level to flag cells with >1 GΩ seal recorded before break-in, initial access resistance <1 or >20 MΩ or >25% of the input resistance. Cell recordings failing these tests were manually examined for recording quality and manually passed or failed. Cells also had to have features successfully extracted for long square pulse sweeps at a minimum to be included in analysis. For mouse V1Sp cells, slightly stricter automated QC values were imposed at the sweep and the cell level, following the original publication.

Transcriptomic data collection

cDNA amplification and library construction

We performed all steps of RNA-processing and sequencing as described in our previous human Patch-seq studies (22, 26, 29, 52). We used the SMART-Seq v4 Ultra Low Input RNA Kit for Sequencing (Takara, 634894) to reverse transcribe poly(A) RNA and amplify full-length cDNA according to the manufacturer's instructions. We performed reverse transcription and cDNA amplification for 20 PCR cycles in 0.65-ml tubes, in sets of 88 tubes at a time. At least 1 control 8-strip was used per amplification set, which contained four wells without cells and four wells with 10 pg control RNA. Control RNA was either Universal Human RNA (UHR) (Takara 636538) or control RNA provided in the SMART-Seq v4 kit. All samples proceeded through Nextera XT DNA Library Preparation (Illumina FC-131-1096) using either Nextera XT Index Kit V2 Sets A to D (FC-131-2001,2002,2003,2004) or custom dual-indexes provided by Integrated DNA 5.

Technologies (IDT). Nextera XT DNA Library prep was performed according to manufacturer's instructions except that the volumes of

all reagents including cDNA input were decreased to 0.2× by volume. Each sample was sequenced to approximately 1 million reads.

RNA-seq data processing

Fifty-base-pair paired-end reads were aligned to GRCh38.p2 using a RefSeq annotation gff file retrieved from NCBI on 11 December 2015 for human and to GRCm38 (mm10) using a RefSeq annotation gff file retrieved from NCBI on 18 January 2016 for mouse (https://www.ncbi.nlm.nih.gov/genome/annotation_euk/all/). Sequence alignment was performed using STAR (v2.5.3, RRID:SCR_004463) (74) in two-pass Mode. PCR duplicates were masked and removed using STAR option bamRemoveDuplicates. Only uniquely aligned reads were used for gene quantification. Gene counts were computed using the R Genomic Alignments package summarizeOverlaps function using Intersection-NotEmpty mode for exonic and intronic regions separately (75). Expression levels were calculated as counts of exonic plus intronic reads. For most analyses, $\log_2(\text{counts per million (CPM)} + 1)$ -transformed values were used, or CPM in the case of Seurat or d' analyses.

Anatomical annotations

Layer annotation and alignment

To characterize the position of biocytin-labeled cells, a 20× brightfield and fluorescent image of DAPI (4',6-diamidino-2-phenylindole) stained tissue was captured and analyzed to determine layer position (human and mouse) and region (mouse only). Using the brightfield and DAPI image, soma position and laminar borders were manually drawn for all human neurons and reconstructed mouse cells and were used to calculate depth relative to the pia, white matter, and/or laminar boundaries. The border between L1 and L2 in particular was visually identified as a characteristic sharp increase in cell-body size and density as well as the presence of clear pyramidal-shaped cell bodies in L2. Laminar locations were calculated by finding the path connecting pia and white matter that passed through the cell's soma coordinate and measuring distance along this path to laminar boundaries, pia, and white matter. For mouse cells without reconstructions, pia and white matter boundaries from the CCF were used as references, with layer positions calculated by aligning the relative cortical depth to an average set of layer thicknesses. Because laminar borders are not hard boundaries obeyed by cell types or the morphoelectric properties of those cell types, we elected to include border cells (cells within 40 μm of the border) in the analysis of Patch-seq data.

For reconstructed neurons, laminar depths were calculated for all segments of the morphology, and these depths were used to create a "layer-aligned" morphology by first rotating

the pia-to-WM axis to vertical, then projecting the normalized laminar depth of each segment onto an average cortical-layer template.

CCF pinning and alignment

Mouse cells were individually manually placed in the appropriate cortical region and layer within the Allen Mouse Common Coordinate Framework (CCF) (76) by matching the 20× image of the slice with a "virtual" slice at an appropriate location and orientation within the CCF.

Human brain region pinning

Available surgical photodocumentation [magnetic resonance imaging (MRI) or brain model annotation] is used to place the human tissue blocks in approximate 3D space by matching the photodocumentation to a MRI reference brain volume "ICBM 2009b Nonlinear Symmetric" (77), with Human CCF overlaid (78) within the ITK-SNAP interactive software.

Morphological reconstruction

Biocytin histology

A horseradish peroxidase (HRP) enzyme reaction using diaminobenzidine (DAB) as the chromogen was used to visualize the filled cells after electrophysiological recording, and DAPI stain was used to identify cortical layers as described previously (29).

Imaging of biocytin-labeled neurons

Mounted sections were imaged as described previously (79). In brief, operators captured images on an upright AxioImager Z2 microscope (Zeiss, Germany) equipped with an Axiocam 506 monochrome camera and 0.63× Optivar lens. Two-dimensional (2D) tiled overview images were captured with a 20× objective lens (Zeiss Plan-NEOFLUAR 20×/0.5) in brightfield transmission and fluorescence channels. Tiled image stacks of individual cells were acquired at higher resolution in the transmission channel only for the purpose of automated and manual reconstruction. Light was transmitted using an oil-immersion condenser (1.4 NA). High-resolution stacks were captured with a 63× objective lens (Zeiss Plan-Apochromat 63×/1.4 Oil or Zeiss LD LCI Plan-Apochromat 63×/1.2 Imm Corr) at an interval of 0.28 μm (1.4 NA objective) or 0.44 μm (1.2 NA objective) along the *z* axis. Tiled images were stitched in ZEN software (RRID:SCR_013672) and exported as single-plane TIFF files.

Morphological reconstruction

Reconstructions of the dendrites and the full axon were generated for a subset of neurons with good-quality transcriptomics, electrophysiology, and biocytin fill. Reconstructions were generated based on a 3D image stack that was run through a Vaa3D-based image processing

and reconstruction pipeline (80). For some cells, images were used to generate an automated reconstruction of the neuron using TReMAP (81). Alternatively, initial reconstructions were created manually using the reconstruction software PyKNOSOS (<https://www.ariadne.ai/>) or the citizen neuroscience game Mozak (82) (<https://www.mozak.science>). Automated or manually initiated reconstructions were then extensively manually corrected and curated using a range of tools (for example, virtual finger and polyline) in the Mozak extension (Zoran Popovic, Center for Game Science, University of Washington) of Terafly tools (83, 84) in Vaa3D. Every attempt was made to generate a completely connected neuronal structure while remaining faithful to image data. If axonal processes could not be traced back to the main structure of the neuron, they were left unconnected.

Slice immunohistochemistry

Immunohistochemistry and slide imaging

Tissue slices (350 μm-thick) designated for histological profiling were fixed for 2 to 4 days in 4% paraformaldehyde (PFA) in phosphate-buffered saline (PBS) at 4°C and transferred to PBS, 0.1% sodium azide for storage at 4°C. For human samples, these slices were interspersed with Patch-seq slices when preparing each tissue block, while for mouse the histology slices were from separate tissue blocks prepared following the same protocol. Slices were then cryoprotected in 30% sucrose, frozen and resectioned at 30 μm (human tissue) and 20 μm (mouse tissue) using a sliding microtome (Leica SM2000R). Sections were stored in PBS with azide at 4°C in preparation for immunohistochemical staining. Staining for Neu-N (Neuronal nuclei; Millipore, MAB377, 1:2,000) with DAB was applied using the Biocare Intellipath FLX slide staining automated platform [as previously in (29)]. The images of stained subsections were acquired with 20× objective on Aperio microscope at a resolution of 1 μm to 1 pixel (human slices) and 1 μm to 0.989 (mouse slices). Full immunohistology protocol details available at http://help.brain-map.org/download/attachments/8323525/CellTypes_Morph_Overview.pdf?version=4&modificationDate=1528310097913&api=v2

Cytoarchitecture analysis

Images of Neu-N stained sections in which somas were clearly distinguishable were selected for further analysis [human MTG: *N*=76 regions of interests (ROIs), 3016 cells, 19 slices from 16 cases, FCtx: *N*=23 ROIs, 1113 cells, 8 slices from 8 cases; TeA: *N*=29 ROIs, 400 cells, 8 slices, 3 animals; VISP: *N*=29 ROIs, 425 cells, 8 slices, 3 animals]. The quantification of cell densities and cell body sizes was performed using custom MATLAB scripts (R2022a, Mathworks, RRID:SCR_001622). Within each subsection, several ROIs containing only L1 were

selected manually, each ROI a rectangle of 500 to 700 μm in length and the full extent of L1 in height. The border between L1 and L2 was visually identified as a characteristic sharp increase in cell-body size and density as well as the presence of clear pyramidal-shaped cell bodies in L2. The MATLAB image-processing scripts identified the cells by binarizing the image and applying watershed transform. Cell area was extracted after applying the scaling factor (1 $\mu\text{m}/\text{pixel}$ for human and 0.989 $\mu\text{m}/\text{pixel}$ for mouse). The cell densities were calculated as the number of cells divided by the volume of tissue (the product of selected ROI area and tissue subsection thickness: 30 μm for human tissue and 20 μm for mouse tissue). Statistical tests were performed using Kruskal Wallis test with post hoc comparisons.

Transcriptomic data analysis

Reference data from dissociated cells and nuclei

Reference transcriptomic data used in this study were obtained from dissociated inhibitory cells (mouse) or nuclei (human) collected from human MTG (21) and mouse VISp (46) and are publicly accessible at the Allen Brain Map data portal (<https://portal.brain-map.org/atlas-and-data/rnaseq>) and Transcriptomics Explorer (RRID:SCR_017567). Layer 1 t-types were assessed by proportions in these datasets after first reducing sampling bias by selecting only samples with even dissections across all cortical layers and in mouse additionally restricting to samples targeted by pan-neuronal or pan-GABAergic mouse lines. As previously described (21), 500- μm thick sections were stained for Nissl to permit visualization of layers. Layer dissections were made based on Nissl stain under a dissecting microscope using a needle blade microknife.

Definition of L1 types from reference data

To accommodate imprecision in L1 dissections, we used a high cutoff for proportion of type in L1 to be considered a L1 type. L1 t-types were defined as types making up >5% of L1 cells or with >50% of the type found in L1 dissections. For human t-types (with relatively unbiased Patch-seq sampling), we verified borderline t-types with the more-precise layer boundaries in Patch-seq data, excluding the VIP LBH type (<1% of L1, <25% in L1) and including the PAX6 TNFAIP8L3 type (>1% of L1, >50% in L1).

For further analysis of these reference transcriptomic datasets, data was restricted to only L1 and the adjacent layer (L2 in human, L2/3 in mouse), to be as selective as possible while not risking the exclusion of L1/2 border cells in case of imprecise dissections.

Discriminant analysis (d')

For measuring distinctness between types in transcriptomic feature space, we followed the

cross-validated negative binomial (NB) discriminant analysis from (23). For each pair of t-types, a set of cross-validated log-likelihood ratios (LLR) were calculated for each cell, fitting a NB classifier to the training split and measuring LLR for the test split across rounds of fivefold cross-validation. The classifier was a naive Bayes negative binomial model, with independent negative binomial distributions fit for each subset on each feature (gene) by maximum likelihood, with dispersion parameter set to $r = 1$ following the observed statistics of our dataset.

This produced a distribution of likelihood ratios for each t-type in the pair, the separation of which was summarized by the d' statistic. For normal distributions this is typically calculated as the separation of means divided by the standard deviation, but we instead used a nonparametric form (equivalent in the normal distribution case): $d' = 2\Phi - 1(\text{AUC})$, where Φ is the CDF of the standard normal distribution, and AUC is the area under the receiver operating characteristic curve for the classifier (equivalently, the proportion of pairs selected one from each type for which the LLR of the cluster 1 cell is higher than the cluster 2 cell).

This method was adapted for other modalities (Patch-seq transcriptomics, morphology, and electrophysiology) by simply using different classifiers. For electrophysiology and morphology, we used a random forest classifier with scikit-learn default parameters and balanced class weighting. For Patch-seq transcriptomics, we modified the naive Bayes negative binomial model to use a zero-inflated negative binomial distribution (statsmodels) (85). Given the high number of free parameters in this model, it was not directly suitable to fitting on the small datasets necessary for the pairwise discriminant analysis. Using ZINB fits to each gene across the full reference dataset, we observed that the parameters π (zero-inflation probability) and ϕ (NB dispersion or shape parameter) both nearly followed a curve depending on μ , the NB distribution mean. We parametrized these curves (ϕ with a spline fit, π with a sigmoid), and used them to constrain ZINB fits for discriminant analysis, assuming that for all genes, zero-inflation parameters follow the same dependence on mean expression. Given these constraints, the maximum likelihood fit for each gene could be implemented simply as a lookup table.

Subclasses and cross-species homology

Human L1 transcriptomic subclasses were defined based on the d' values by grouping all pairs of t-types with $d' < 2.2$, equivalent to approximately 2% overlap of LLR distributions. Four pairs of t-types were grouped in this way, forming three subclasses.

We defined cross-species homology of L1 t-types following a variation of the procedure

in (27), using coordinates for each cell in a space integrating the mouse and human transcriptomic references (calculated in the original from ScAlign, 30 dimensions). Instead of relying on defining clusters in that integrated space and measuring overlap within those clusters, we directly defined a similarity metric for any two clusters in that space: the ratio of the mean intracluster difference to the mean intercluster difference. Intracluster difference was averaged over all pairs of cells within each cluster, then averaged over the two clusters; intercluster difference was averaged directly over all intercluster pairs. We summarized this similarity metric for all mouse L1 t-types aligned to human L1 subclasses (Fig. 1) and to individual human L1 t-types (Fig S1).

Patch-seq data curation and mapping

Patch-seq samples were included in this dataset if they met the following transcriptomic quality criteria: a normalized sum of “on” type marker-gene expression (NMS) greater than 0.4 and a normalized sum of nonneuronal contamination markers less than 2 (86).

We mapped Patch-seq samples to reference taxonomies from the reference single cell/nuclei RNA-sequencing datasets introduced above, consisting of a hierarchical dendrogram of cell types, with a subset of cells from the reference identified with each node of the tree and a set of marker genes defined to distinguish types at each split in the tree (up to 50 markers in each direction per branch point, limited to robust markers only as defined by `select_markers_pair_group` in the `scrattch.hicat` package). The Patch-seq transcriptomes were mapped to the reference taxonomy following the “tree mapping” method (`map_dend_membership` in the `scrattch.hicat` package). Briefly, at each branch point of the taxonomy we computed the correlation of the mapped cell’s gene expression with that of the reference cells on each branch, using the markers associated with that branch point (i.e., the genes that best distinguished those groups in the reference) and chose the most correlated branch. The process was repeated until reaching the leaves of the taxonomy (t-types). To determine the confidence of mapping, we applied 100 bootstrapped iterations at each branch point, and in each iteration, 70% of the reference cells and 70% of markers were randomly sampled for mapping. The percentage of times a cell was mapped to a given t-type was defined as the mapping probability, and the highest probability t-type was assigned as the mapped cell type.

Only cells mapping to the identified L1 t-types were included in subsequent morpho-electric feature analysis. Neurons from non-L1 t-types present in human L1 were also included in the L1 proportion analysis, and those for which morphological reconstructions were

available were included in the supplementary morphology gallery. Some additional quality filters were applied to the mouse VIS cells only, following the procedure in their original publication: excluding cells with poor RNA amplification and “inconsistent” cells as defined by unexpected patterns of mapping probabilities.

Joint visualization

We visualized transcriptomic diversity using a nonlinear projection of a transcriptomic space following integration of each species' Patch-seq and reference dissociated cell/nuclei datasets. We first excluded genes potentially related to technical variables: X and Y chromosome genes, mitochondrial genes [Human MitoCarta2.0], and genes most highly expressed in a nonneuronal cell type in the reference dataset. For human Patch-seq samples, which had more variable quality of transcriptomic data, we additionally excluded a small set of immune/glial activation-related genes that were shown to introduce non-cell-type-related variability (65, 87).

This filtered gene set was loaded and processed by the Seurat pipeline (33): Expression values were first normalized by the SCTransform model, then the 3000 most variable genes were transformed by CCA and nonlinear warping to integrate the Patch-seq and reference datasets (functions FindIntegrationAnchors and IntegrateData). For human Patch-seq samples only, the SCTransform normalization additionally reduced effects of contamination by regressing against the normalized contamination marker sum for each cell. The integrated space was then transformed by PCA (30 PCs) followed by UMAP projection (to two dimensions) for visualization.

Electrophysiology feature analysis

For all electrophysiology stimuli that elicited spiking, action potentials were detected by first identifying locations where the smoothed derivative of the membrane potential (dV/dt) exceeded 20 mV ms^{-1} , then refining on the basis of several criteria, including threshold-to-peak voltage, time differences, and absolute peak height. For each action potential, threshold, height, width (at half-height), fast after-hyperpolarization (AHP) and interspike trough were calculated (trough and AHP were measured relative to threshold), along with maximal upstroke and downstroke rates dV/dt and the upstroke/downstroke ratio (that is, ratio of the peak upstroke to peak downstroke).

Following spike detection, summary features were calculated from sweeps with long square pulse current injection: input resistance (all hyperpolarizing sweeps, -10 to -90 pA), sag (hyperpolarizing sweep with response closest to -100 mV , generally -90 pA stimulus, and

depolarizing sag on subthreshold response closest to rheobase), rheobase, and f-I slope (all spiking sweeps, up to rheobase $+80 \text{ pA}$). Spike-train properties were calculated for each spiking sweep: latency, average firing rate, initial instantaneous firing rate (inverse of first ISI), mean and median ISI, ISI CV, irregularity ratio, and adaptation index. These spike-train features and the single-spike properties listed above (measured on the first action potential) were summarized for both the rheobase sweep and a stimulus 40 pA above rheobase. For spike upstroke, downstroke, width, threshold, and interspike interval (ISI), “adaptation ratio” features were calculated as a ratio of the spike features between the first and third spike (on the lowest amplitude stimulus to elicit at least four spikes).

Spike-shape properties were also calculated for short (3-ms) pulse stimulation and a slowly increasing current ramp stimulus (first spike only). A subset of cells also had subthreshold frequency response characterized by a logarithmic chirp stimulus (sine wave with exponentially increasing frequency), for which the impedance profile was calculated and characterized by features including the peak frequency and peak ratio. Feature extraction was implemented using the IPFX python package (); custom code used for chirps and some high-level features will be released in a future version of IPFX.

Morphology feature analysis

Prior to morphological feature analysis, reconstructed neuronal morphologies were expanded in the dimension perpendicular to the cut surface to correct for shrinkage (88, 89) after tissue processing. The amount of shrinkage was calculated by comparing the distance of the soma to the cut surface during recording and after fixation and reconstruction. For mouse cells, a tilt-angle correction was also performed based on the estimated difference (via CCF registration) between the slicing angle and the direct pia-white matter direction at the cell's location (79). Features predominantly determined by differences in the z -dimension were not analyzed to minimize technical artifacts due to z -compression of the slice after processing.

Morphological features were calculated as previously described (79). In brief, feature definitions were collected from prior studies (90, 97). Features were calculated using the skeleton keys (https://github.com/AllenInstitute/skeleton_keys) and neuron morphology (https://github.com/AllenInstitute/neuron_morphology) python packages. Features were extracted from neurons aligned in the direction perpendicular to pia and white matter. Laminar axon distribution (bin size of $5 \mu\text{m}$) and earth movers distance features require a layer-aligned version of the morphology where node depths

are registered to an average interlaminar depth template.

Morphoelectric modeling analysis

To explore relationships between morphology and input resistance, biophysically detailed multicompartmental models were tested for all LAMP5 cells in both species using the BMTK modeling framework (92) and the NEURON (RRID:SCR_005393) simulator. Models were instantiated for each morphology using a single set of parameters (leak and intracellular conductance, membrane resistance, ion channels, etc.) obtained from a previously optimized model (79) for a mouse NGFC cell (viewable at <http://celltypes.brain-map.org/experiment/electrophysiology/475585413>). After an initial transient to ensure models were stable at their resting membrane potential, input resistance was measured using the NEURON Impedance tool for linearized input resistance at the soma compartment. The resulting morphology-predicted input resistance values were regressed against measured values by species to assess how strongly morphologies contributed to observed variation in input resistance.

Statistical analysis of variability

Unless otherwise specified, statistical analyses were implemented in python using the statsmodels package (85), and clustering and classification methods were implemented using scikit-learn (93). Samples consisted of independent biological replicates (Patch-seq recorded neurons), with counts given in table S2.

Variation by subclass and species

To assess the variability of morphoelectric features by subclass within species, we used a one-way analysis of variance (ANOVA) on ranks [Kruskal-Wallis (KW) test] for each feature by subclass. Results were reported as fraction of variance explained (ϵ^2) and KW test P value. P values were corrected for false discovery rate (FDR, Benjamini-Hochberg procedure) across all features for each data modality. Post-hoc Dunn's tests were run across all pairs of subclasses (excluding ungrouped t-types), and results were FDR-corrected. Analysis of feature relationships with other variables including cell depth, brain region, or donor characteristics were likewise assessed by Mann-Whitney (MW) tests for binary variables, KW test for categorical, and Spearman's or Pearson's correlations for continuous variables, all FDR-corrected across features by modality.

For cross-species analysis, samples were restricted to only cells present within L1 and belonging to one of the homologous subclasses in each species. Overall cross-species variation was assessed by a MW test for each feature, ranked by effect size r , and FDR-corrected by modality. Subclass-dependence of these differences were assessed by a two-way ANOVA

on species and subclass (heteroskedasticity-corrected); where species-subclass interactions were found along with species differences, this was followed by post-hoc tests for species differences within each subclass (MW test).

Clustering and classification

Clustering and classification tasks required some preprocessing of electrophysiology data to deal with missing and outlier values: We permitted cells with partially incomplete recordings, for instance, to maximize the usage of available data. Preprocessing included outlier removal, data standardization, and imputation, after excluding cells with more than 60% of electrophysiological features missing. Extreme outliers were removed first (LocalOutlierFactor < -20); for standardization, features were centered about the median and scaled by interquartile range (IQR) (RobustScaler); missing values were imputed as the mean of five nearest neighbors (KNNImputer).

Following this preprocessing, electrophysiology and morphology classifiers were trained and tested in a pairwise manner, following the discriminant analysis technique described above, as well as on the full multiclass problem of assigning subclass labels to the full dataset based on electrophysiology. For this problem a multiclass logistic regression classifier was used, with balanced class weights. To assess within-dataset performance, repeated stratified fivefold cross-validation was used, with classifier predictions on test data aggregated across cross-validation folds to calculate a confusion matrix of performance. Performance was additionally assessed on a held-out secondary electrophysiology dataset, as described in the main text. To prevent features with dataset dependence from degrading performance, affected features were excluded based on a one-way ANOVA for the effects of dataset (primary or two secondary datasets collected at different sites). Features with $P < 0.05$ or $R^2 > 0.05$ were excluded.

To demonstrate and visualize discrimination based on small subsets of electrophysiology features, we searched for 1D- and 2D- feature subspaces in which each subclass clustered separately from all other cells. A two-cluster Gaussian mixture model was fit to the data in each subspace, and performance was assessed by f1 score (harmonic mean of precision and recall) after identifying the cluster that best matched the subset of interest. Results were shown for the highest-ranked subspace for each subclass.

MERFISH data collection

Human postmortem frozen brain tissue was embedded in Optimum Cutting Temperature medium (VWR,25608-930) and sectioned on a Leica cryostat at -17°C at $10\text{ }\mu\text{m}$ onto Vizgen MERSCOPE coverslips. These sections were then processed for MERSCOPE imaging according to the manufacturer's instructions.

Briefly, sections were allowed to adhere to these coverslips at room temperature for 10 min prior to a 1-min wash in nuclease-free phosphate buffered saline (PBS) and fixation for 15 min in 4% paraformaldehyde in PBS. Fixation was followed by 3×5 min washes in PBS prior to a 1-min wash in 70% ethanol. Fixed sections were then stored in 70% ethanol at 4°C prior to use and for up to 1 month. Human sections were photobleached using a 150-W LED array for 72 hours at 4°C prior to hybridization then washed in 5 ml Sample Prep Wash Buffer (VIZGEN 20300001) in a 5-cm petri dish. Sections were then incubated in 5 ml Formamide Wash Buffer (VIZGEN 20300002) at 37°C for 30 min. Sections were hybridized by placing 50 μl of VIZGEN-supplied Gene Panel Mix onto the section, covering with parafilm, and incubating at 37°C for 36 to 48 hours in a humidified hybridization oven. Following hybridization, sections were washed twice in 5 ml Formamide Wash Buffer for 30 min at 47°C . Sections were then embedded in acrylamide by polymerizing VIZGEN Embedding Premix (VIZGEN 20300004) according to the manufacturer's instructions. Sections were embedded by inverting sections onto 110 μl of Embedding Premix and 10% Ammonium Persulfate (Sigma A3678) and TEMED (BioRad 161-0800) solution applied to a Gel Slick (Lonza 50640) treated 2×3 glass slide. The coverslips were pressed gently onto the acrylamide solution and allowed to polymerize for 1.5 hours. Following embedding, sections were cleared for 24 to 48 hours with a mixture of VIZGEN Clearing Solution (VIZGEN 20300003) and Proteinase K (New England Biolabs P8107S) according to the manufacturer's instructions. Following clearing, sections were washed twice for 5 min in Sample Prep Wash Buffer (PN 20300001). VIZGEN DAPI and PolyT Stain (PN 20300021) were applied to each section for 15 min followed by a 10-min wash in Formamide Wash Buffer. Formamide Wash Buffer was removed and replaced with Sample Prep Wash Buffer during MERSCOPE setup. 100 μl of RNase Inhibitor (New England BioLabs M0314L) was added to 250 μl of Imaging Buffer Activator (PN 203000015), and this mixture was added via the cartridge activation port to a prethawed and mixed MERSCOPE Imaging cartridge (VIZGEN PN1040004). 15 ml mineral oil (Millipore-Sigma m5904-6X500ML) was added to the activation port and the MERSCOPE fluidics system was primed according to VIZGEN instructions. The flow chamber was assembled with the hybridized and cleared section coverslip according to VIZGEN specifications, and the imaging session was initiated after collection of a 10X mosaic DAPI image and selection of the imaging area. For specimens that passed minimum count threshold, imaging was initiated and processing completed according to VIZGEN proprietary

protocol. Following image processing and segmentation, cells with fewer than 50 transcripts were eliminated, as well as cells with volumes falling outside a range of 100 to $300\text{ }\mu\text{m}$.

Gene panel selection

The 140-gene human cortical panel was selected using a combination of manual and algorithmic based strategies requiring a reference single-cell/nucleus RNA-seq dataset from the same tissue, in this case the human MTG snRNA-seq dataset and resulting taxonomy (2f). First, the reference RNA-seq dataset is filtered to only include genes compatible with mFISH. Retained genes need to be (i) long enough to allow probe design (> 960 base pairs); (ii) expressed highly enough to be detected (FPKM ≥ 10), but not so high as to overcrowd the signal of other genes in a cell (FPKM < 500); (iii) expressed with low expression in off-target cells (FPKM < 50 in nonneuronal cells); and (iv) differentially expressed between cell types [top 500 remaining genes by marker score (94)]. To more evenly sample each cell type, the reference dataset is also filtered to include a maximum of 50 cells per cluster. Second, an initial set of high-confidence marker genes are selected through a combination of literature search and analysis of the reference data.

The main step of gene selection uses a greedy algorithm to iteratively add genes to the initial set. To do this, each cell in the filtered reference dataset is mapped to a cell type by taking the Pearson correlation of its expression levels with each cluster median using the initial gene set of size n , and the cluster corresponding to the maximum value is defined as the "mapped cluster." The "mapping distance" is then defined as the average cluster distance between the mapped cluster and the originally assigned cluster for each cell. In this case, a weighted cluster distance, defined as one minus the Pearson correlation between cluster medians calculated across all filtered genes, is used to penalize cases where cells are mapped to very different types; but an unweighted distance, defined as the fraction of cells that do not map to their assigned cluster, could also be used. This mapping step is repeated for every possible $n+1$ gene set in the filtered reference dataset, and the set with minimum cluster distance is retained as the new gene set. These steps are repeated with the new gene set (of size $n+1$) until a gene panel of the desired size is attained. Code for reproducing this gene selection strategy is available as part of the mfishtools R library (<https://github.com/AllenInstitute/mfishtools>).

Mapping transcriptomic types and calculating proportions

Any genes not matched across both the MERSCOPE gene panel and the mapping taxonomy were filtered from the dataset before

starting. From there, cluster means were calculated by dividing the number of cells per cluster by the number of clusters collected. Next, we created a training dataset by finding marker genes for each cluster by calculating the Euclidean distance between all clusters and the mean counts of each gene per cluster. This training dataset was fed into a KNN classifier alongside the MERSCOPEs cell by gene panel to iteratively calculate the best possible gene matches per cluster. Proportions of L1 t-types present in L1 were calculated in samples from five donors by manually drawing L1 borders and selecting the corresponding cells.

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SUPPLEMENTARY MATERIALS

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Figs. S1 to S9

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